



Lab5 Students' Learning Level Prediction

ENGG1910 Demystifying AI





Introduction

Background information and
setup of Colab

Supporting projects

- Teaching Development and Learning Enhancement Grant: An AI-enhanced Adaptive and Individualized eLearning System for Mathematics Foundation Courses at the Faculty of Engineering

Principal Supervisors



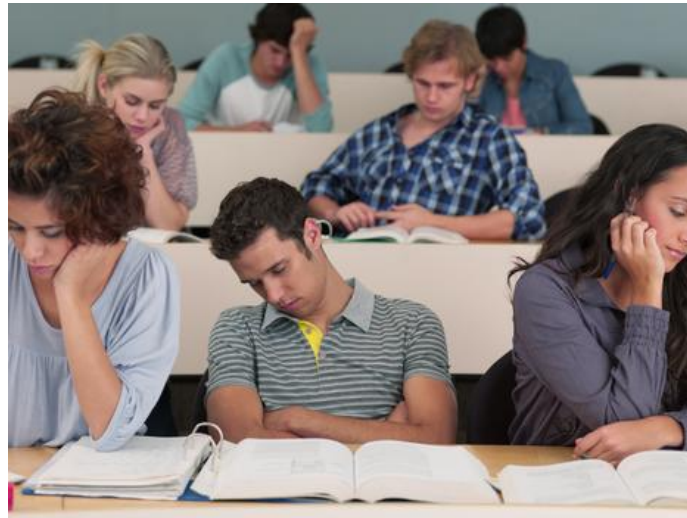
Prof. Tan LEE



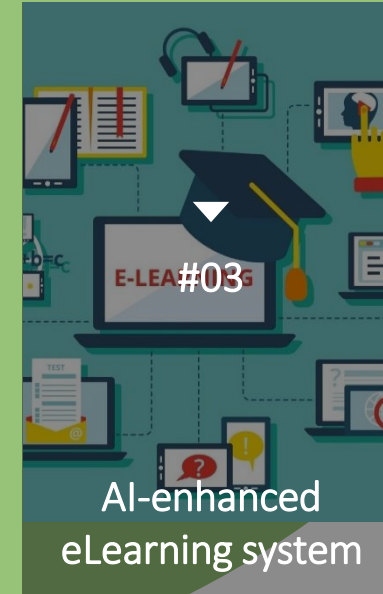
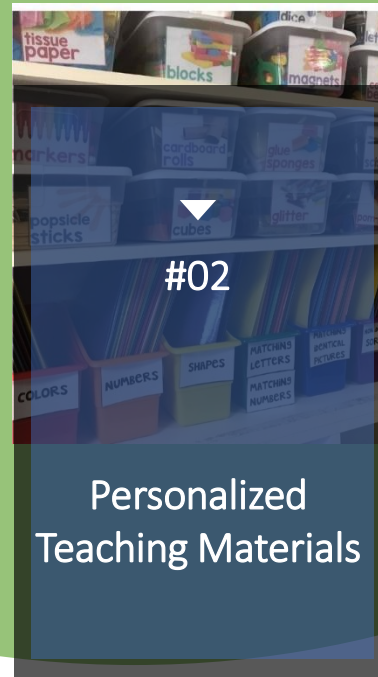
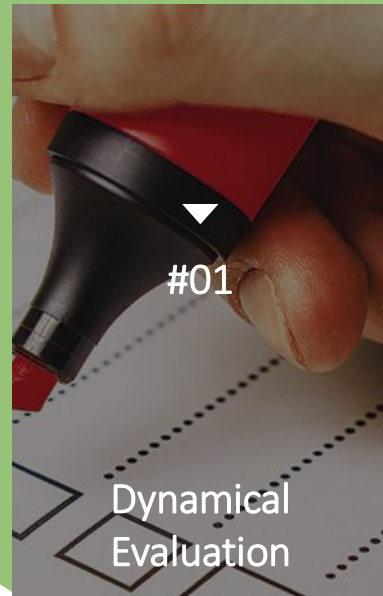
Dr. Dongkun HAN

Main motivations

- Students are different (**variety of learners**)
- Learning materials are different (**variety of knowledge and skills**)
- It is hard to provide suitable knowledge and training to suitable students at a certain learning level (**proper individualized education**)

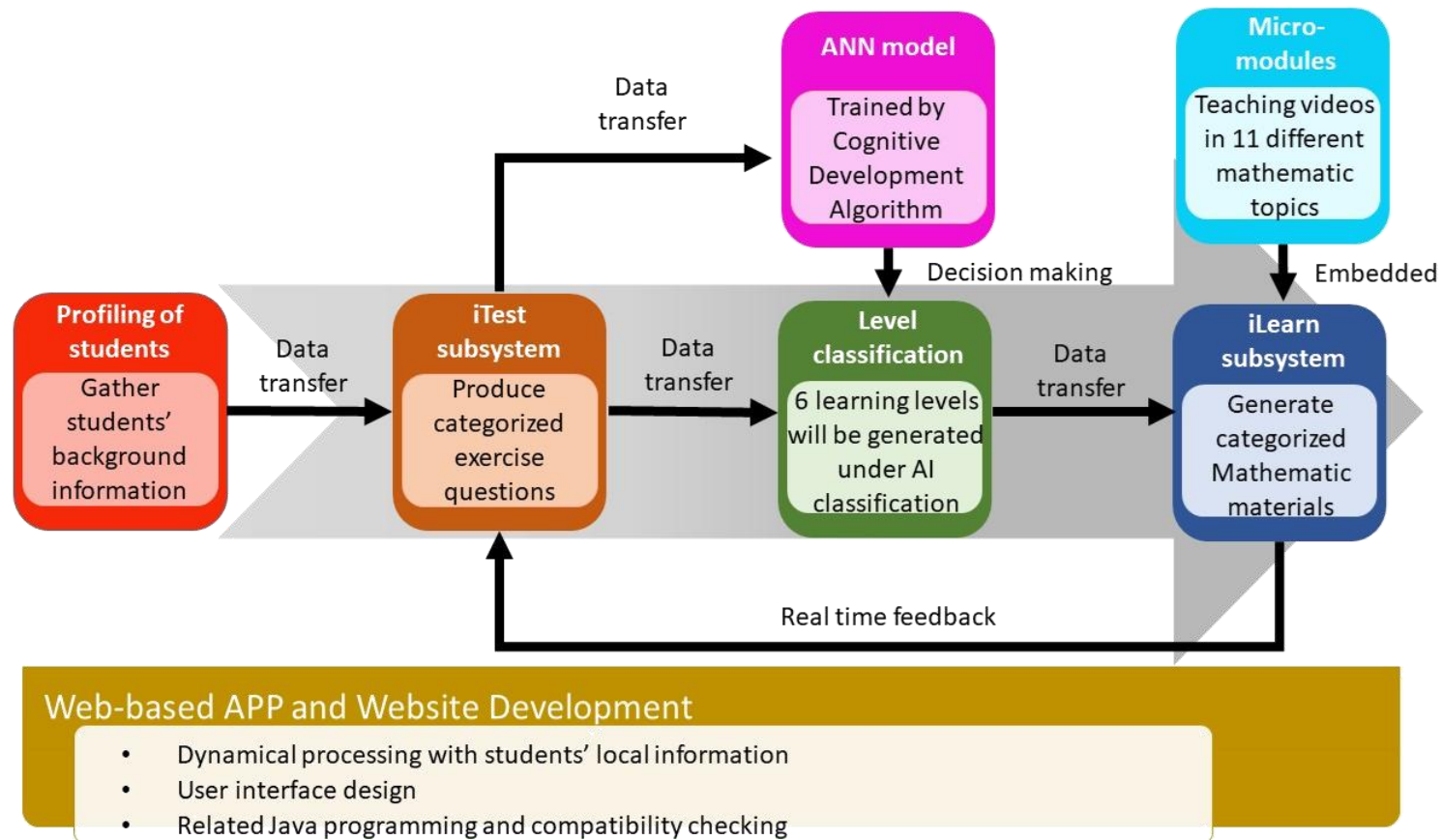


Targets of the Project

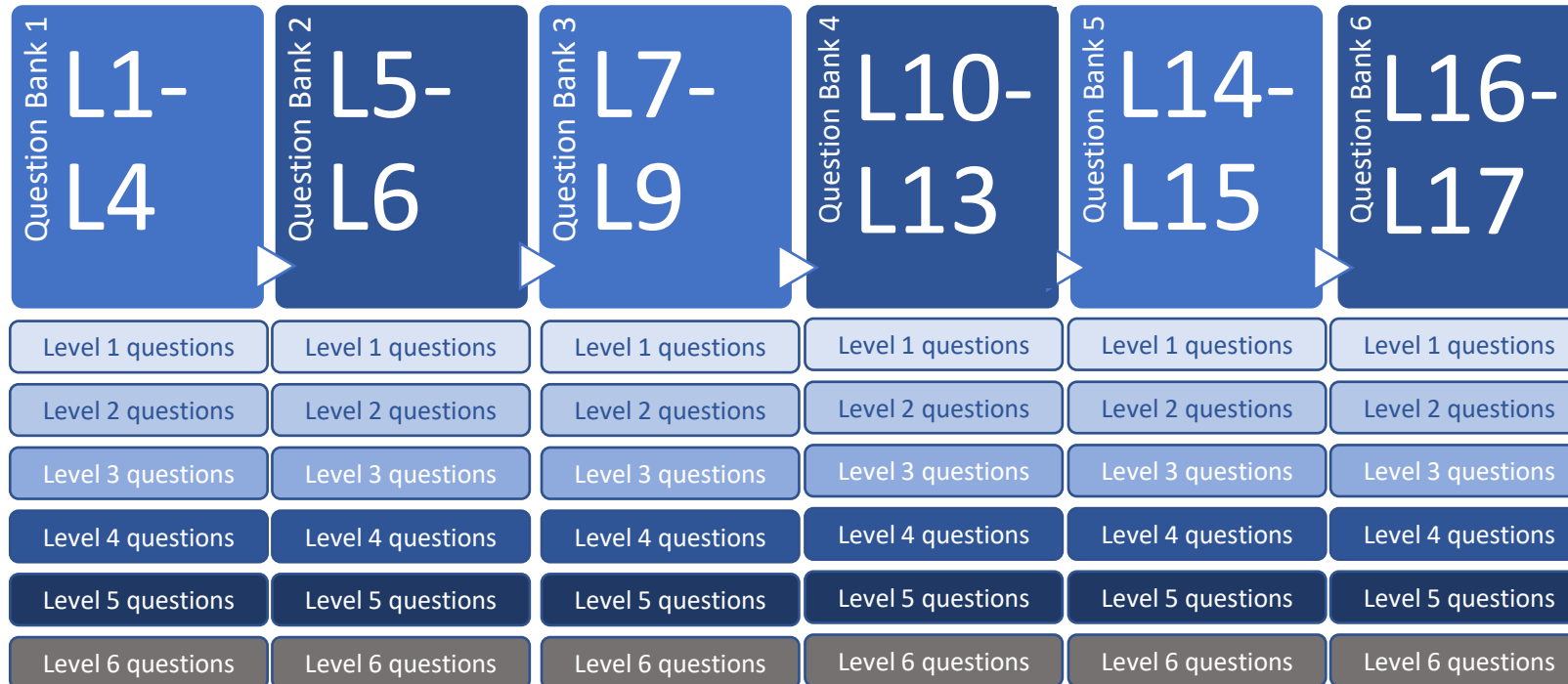


- **Dynamically evaluate** students on different contents in learning mathematics
- Supply personalized and appropriate **teaching materials**
- Develop an **AI-enhanced eLearning system** in teaching mathematic foundation courses

Structure of eLearning system



iTest question bank of ENGG2020



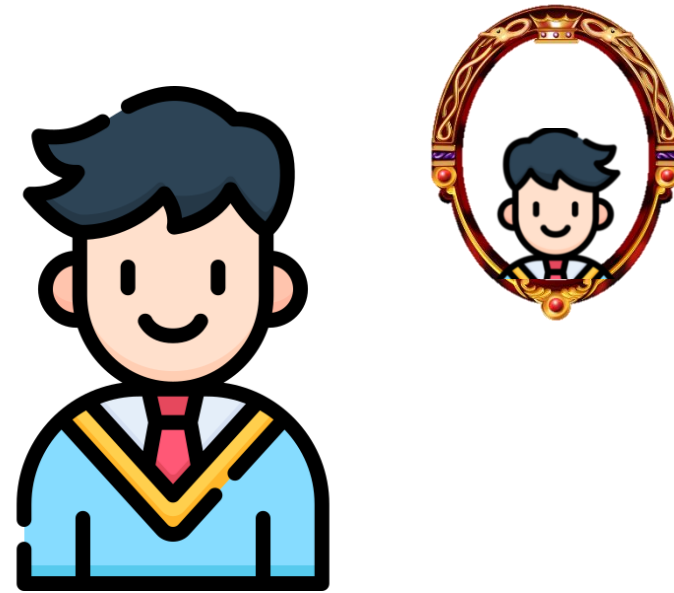
So tired



Prediction is important



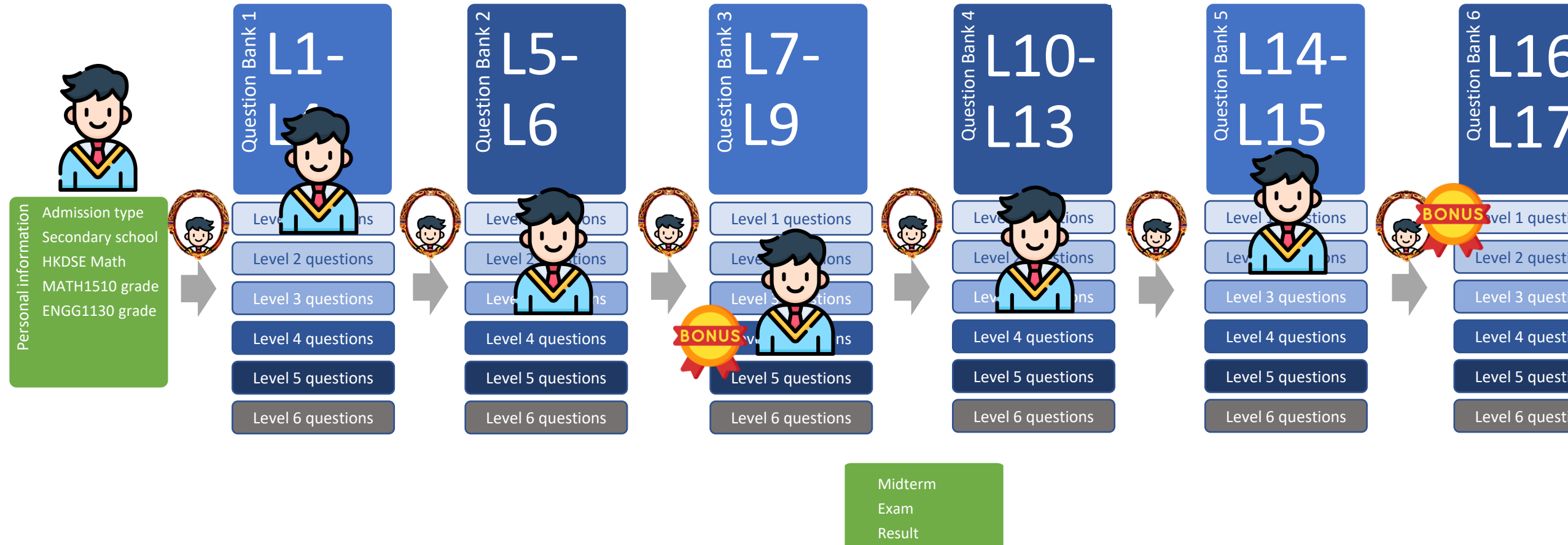
"Mirror, mirror on the wall, Who is the fairest of us all?"



"Mirror, mirror on the wall, Who will get a 'F'?"

"Mirror, mirror on the wall, what is my grade of this course?"

Dynamical evaluation

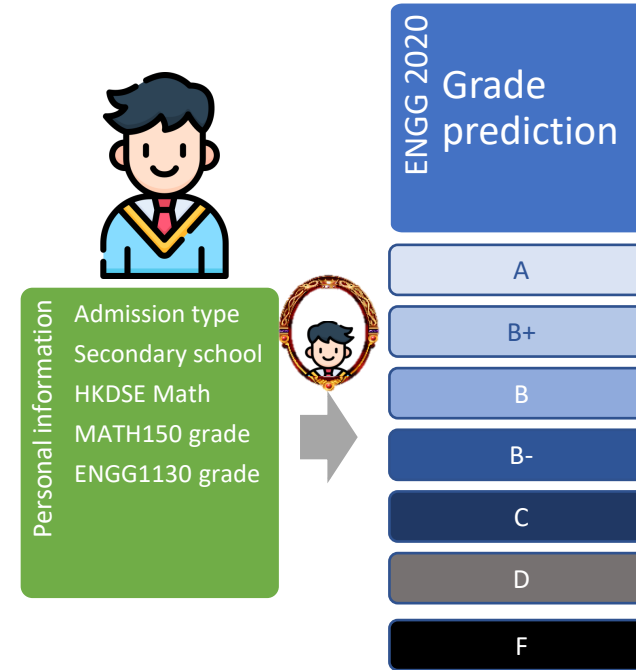


Possible benefits

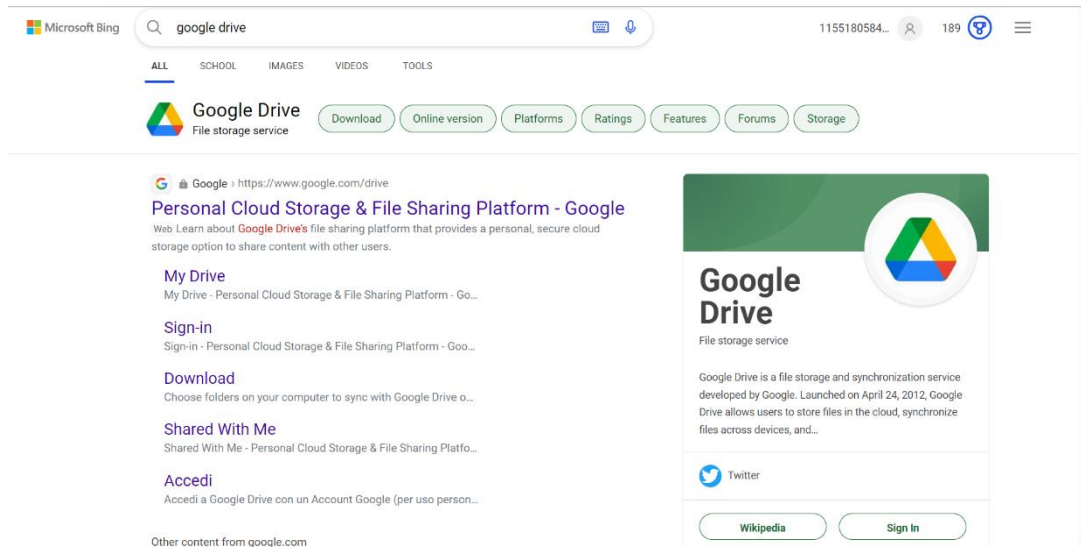
- **Motivations** for self-learning (boosted learning outcome)
- Automatic warning and intervention system (less F)
- Big data analysis for other generative AI tools (digital resources)

Objectives of this lab

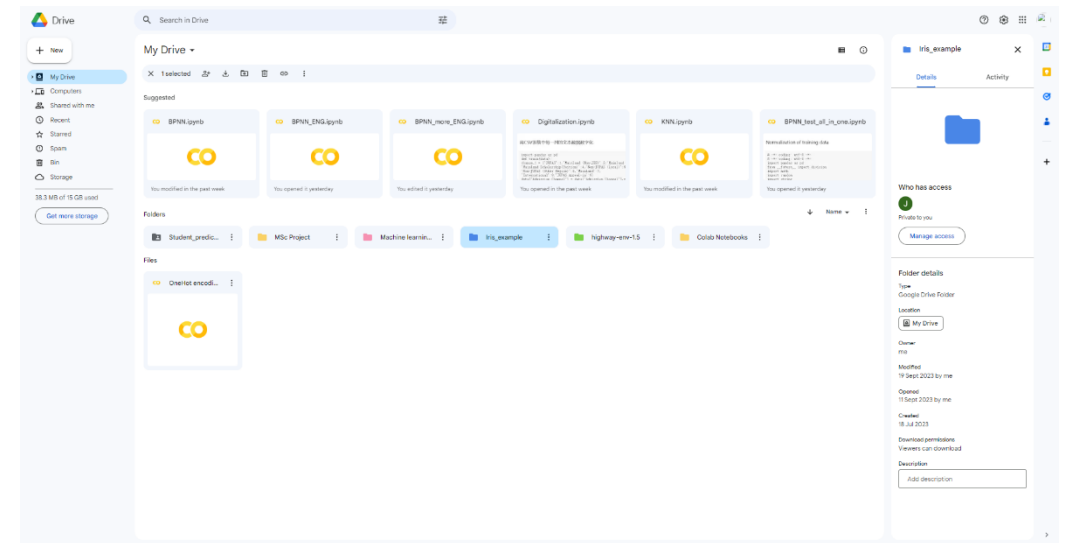
- Setup a programming **platform** for learner level prediction
- Introduce **4 methods** for learner level prediction
- Implement 4 methods on Colab with given data and codes
- Select an optimal method and parameters for your own case



Register a google drive



Register a Google drive: <https://drive.google.com/>



Google drive interface

Download files

- Download the files from Blackboard:

Course Contents/Laboratories/Lab 5 Student Learning Level Prediction

BPNN	11/1/2024 11:15 AM	File folder
Comparison	11/1/2024 11:15 AM	File folder
KNN	11/3/2024 6:13 PM	File folder
Main_code	11/1/2024 11:15 AM	File folder
NB	11/1/2024 11:15 AM	File folder
SVC	11/1/2024 11:15 AM	File folder

Data explanation

- Real information of 2019-2021 undergraduate students at Faculty of Engineering.
- The **features** include Admission channel, Secondary school, Admission exam, Math background, Grade of MATH1510, ENGG1130, ESTR1006 and ENGG2020.

	A	B	C	D	E	F	
1	No	Admission Channel	Secondary School	Exam	Math Background	MATH1510 Grade (2020-21 T1)	EN
2	373	International	#N/A	Kazakhstan High School	Kazakhstan High School - 5/5 in all Math subjects	B+	
3	374	JUPAS	LOK SIN TONG KU CHIU MAN SEC SCH	HKDSE	HKDSE Math: 5	B	
4	375	JUPAS	TWGH WONG FUT NAM COLLEGE	HKDSE	HKDSE Math: 5	B-	
5	376	Mainland	#N/A	Mainland JEE	Mainland JEE Math: 136.5	B+	
6	377	Mainland	#N/A	Mainland JEE	Mainland JEE Math: 147	A-	
7	378	International	#N/A	SAT + SAT Subject Test	SAT Math Lv. 2: 800/800	A-	
8	379	International	#N/A	SAT + SAT Subject Test	SAT Math Lv. 2: 800/800	B-	
9	380	JUPAS	KOWLOON SAM YUK SECONDARY SCHOOL	HKDSE	HKDSE Math: 4, M1: 4	A-	
10	381	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5	P	
11	382	JUPAS	HKWMA CHU SHEK LUN SECONDARY SCHOOL	HKDSE	HKDSE Math: 5, M2: 3	C-	
12	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE Math: 4, M2: 3	C-	
13	384	JUPAS	TWGH CHEN ZAO MEN COLLEGE	HKDSE	HKDSE Math: 4	F	
14	385	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5*, M1: 5*	A-	
15	386	JUPAS	KING LING COLLEGE	HKDSE	HKDSE Math: 5, M2: 4	B	
16	387	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE Math: 5*, M2: 4	C+	
17	388	JUPAS	KING'S COLLEGE	HKDSE	HKDSE Math: 5*, M2: 5	B	
18	389	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 4, M2: Abs	U	
19	390	JUPAS	POOI TO MIDDLE SCHOOL	HKDSE	HKDSE Math: 5, M2: 3	P	
20	391	JUPAS	SKH BISHOP BAKER SECONDARY SCHOOL	HKDSE	HKDSE Math: 5, M1: 3	B-	
21	392	JUPAS	CARMEL SECONDARY SCHOOL	HKDSE	HKDSE Math: 5	B	
22	393	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5, M2: 4	A-	
23	394	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE Math: 5**, M2: 4	B+	
24	395	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 4	P	
25	396	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE Math: 5*, M2: 4	B-	
26	397	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE Math: 5*, M2: 4	B	
27	398	JUPAS	STFA LEE SHAU KEE COLLEGE	HKDSE	HKDSE Math: 5	F	
28	399	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE Math: 4, M2: 4	B-	
29	400	JUPAS	CCC HEIP WOH COLLEGE	HKDSE	HKDSE Math: 4	P	

Data preparation

- Only keep the data of students admitted by HKDSE and Gaokao.
- Only keep the data of students without selecting P/F option.

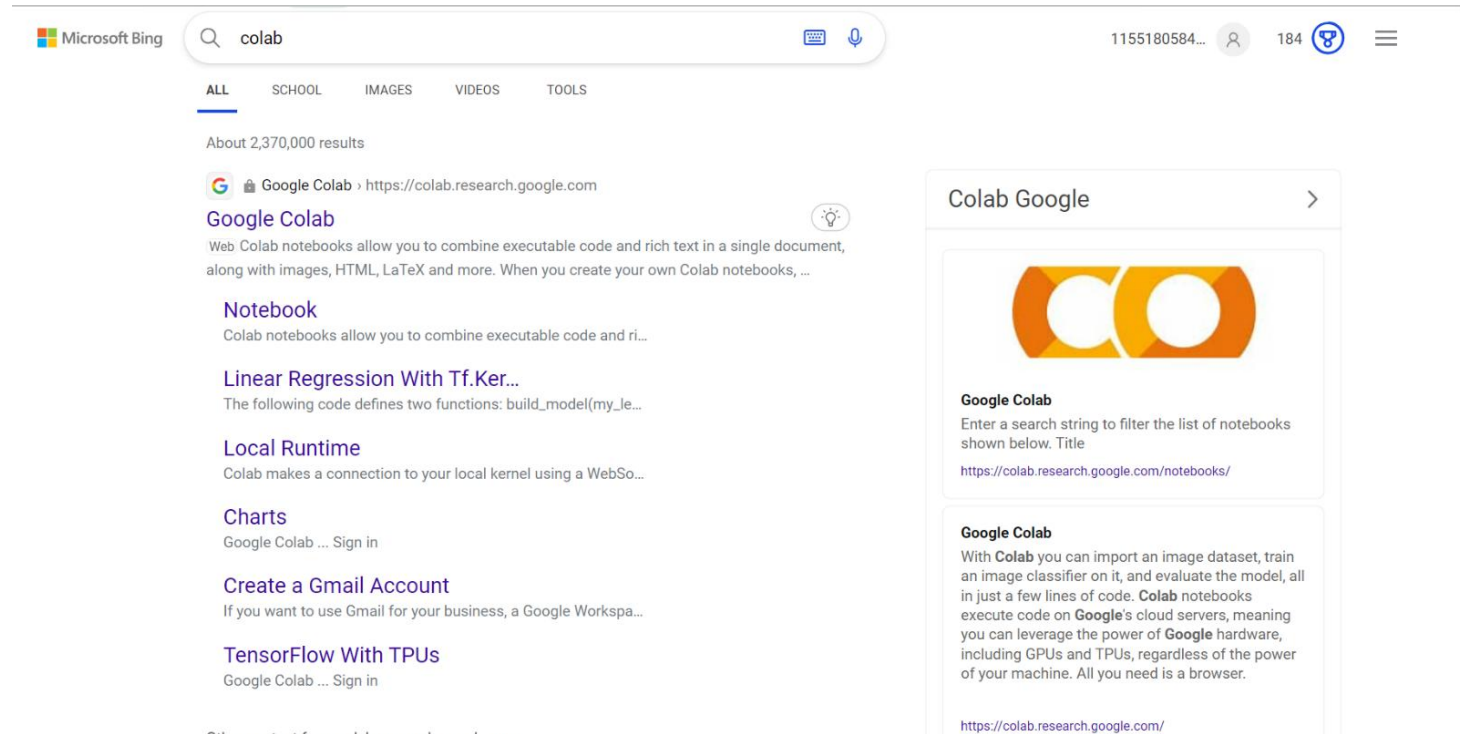
#	A	B	C	D	E	F	G	H
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENG1130	Level
2	17 JUPAS		TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE 5	B+	B+	B+
3	20 JUPAS		CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE 5+	A-	B	B
4	25 JUPAS		MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE 4	A-	A	A
5	29 JUPAS		PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE 4	B+	B-	B-
6	36 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE 5	B-	B+	B+
7	37 JUPAS		METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE 5	D	D+	D
8	50 JUPAS		Others	HKDSE	HKDSE 5	B-	C+	C
9	55 JUPAS		KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE 5	B	B	B
10	62 JUPAS		YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE 4	B	B+	B+
11	82 JUPAS		CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE 5	B	F	F
12	88 JUPAS		HKMA K S LO COLLEGE	HKDSE	HKDSE 4	C	C+	C
13	92 JUPAS		LIONS COLLEGE	HKDSE	HKDSE 4	B	C-	C
14	94 JUPAS		YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE 5**	B+	A-	A
15	98 JUPAS		HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE 5**	B+	B+	B+
16	101 JUPAS		CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE 5	B-	B-	B-
17	109 JUPAS		VALTORTA COLLEGE	HKDSE	HKDSE 5**	A-	A	A
18	119 JUPAS		AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE 5	B-	A-	A
19	129 JUPAS		PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE 5**	A-	A	A
20	130 JUPAS		TWGH LI KA SHING COLLEGE	HKDSE	HKDSE 5	C+	C-	C
21	142 JUPAS		TANG KING PO SCHOOL	HKDSE	HKDSE 4	B+	B-	B-
22	144 JUPAS		PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE 4	B+	B-	B-
23	147 JUPAS	Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE 5	B-	C	C
24	168 JUPAS		FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE 5+	A-	B+	B+
25	172 JUPAS		SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE 4	B	C+	C
26	177 JUPAS		CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE 5	A-	B-	B-
27	182 JUPAS		TSUEN WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5+	B-	D	D
28	183 JUPAS		HONG KONG CHINESE WOMENS CLUB COLLEGE	HKDSE	HKDSE 5	B-	F	F
29	184 JUPAS		TACK CHING GIRLS SECONDARY SCHOOL	HKDSE	HKDSE 4	B	B-	B-
30	187 JUPAS		CHINESE FOUNDATION SECONDARY SCHOOL	HKDSE	HKDSE 4	A-	B	B
31	189 JUPAS		MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE 4	B	A-	A
32	202 JUPAS		SHAU KEI WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5+	B	C-	C
33	208 JUPAS		YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE 4	B-	F	F
34	213 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE 5	A-	A-	A
35	219 JUPAS		SALEM-IMMANUEL LUTHERAN COLLEGE	HKDSE	HKDSE 5+	B+	B	B
36	224 JUPAS		ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE 5+	B-	F	F
37	225 JUPAS		PLK LEE SHING PIK COLLEGE	HKDSE	HKDSE 4	D	C-	C
38	229 JUPAS		Others	HKDSE	HKDSE 5+	A-	A	A
39	232 JUPAS		STFA YUNG YAU COLLEGE	HKDSE	HKDSE 5	D	D+	D
40	240 JUPAS		SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE 5+	A-	A-	A
41	241 JUPAS		PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE 5+	A	A-	A

#	A	B	C	D	E	F	G	H
406	1079 JUPAS		Others	HKDSE	HKDSE 5	C	A-	A
407	1080 JUPAS		TAK OI SECONDARY SCHOOL	HKDSE	HKDSE 5	C	B+	B+
408	1081 JUPAS		SHAU KEI WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5	D+	B	B
409	1082 JUPAS		TSEUNG KWAN O GOVERNMENT SEC SCHOOL	HKDSE	HKDSE 5	A	B+	B+
410	1083 JUPAS		LA SALLE COLLEGE	HKDSE	HKDSE 4	D	B-	B-
411	1086 JUPAS		KIANGSU-CHEKIANG COLLEGE (KWAI CHUNG)	HKDSE	HKDSE 5	C	B-	B-
412	1088 JUPAS		HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE 4	C	B-	B-
413	1090 JUPAS		PENTECOSTAL LAM HON KWONG SCHOOL	HKDSE	HKDSE 5+	B-	A-	A
414	10 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A	A
415	120 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A	A
416	131 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
417	132 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
418	201 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A	A
419	207 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B-	B-
420	212 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A-	A
421	214 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
422	216 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)110-119	A	B-	B-
423	226 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	B+	B+
424	227 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A-	B+	B+
425	234 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A-	A
426	260 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
427	261 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
428	265 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B-	B-
429	269 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A-	A
430	278 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B+	B+
431	281 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A-	B+	B+
432	287 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A-	A
433	303 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B+	B+
434	304 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B+	B+
435	305 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A-	A
436	306 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	A-	A
437	307 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	B-	B-
438	325 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	B-	B-
439	327 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
440	329 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B-	B-
441	330 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	B+	A	A
442	331 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	B+	B+
443	351 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	C	C
444	355 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	A	A
445	357 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
446	365 Mainland		Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	B+	B+

Manipulated data

Visit and log-in Colab

- Search “colab” in your browser (<https://drive.google.com/>).
- Click the first search result.
- Sign in with your Google account if there is one.



Log-in with your google account



4 methods

Mechanism and implementations on
learner level prediction

Method overview

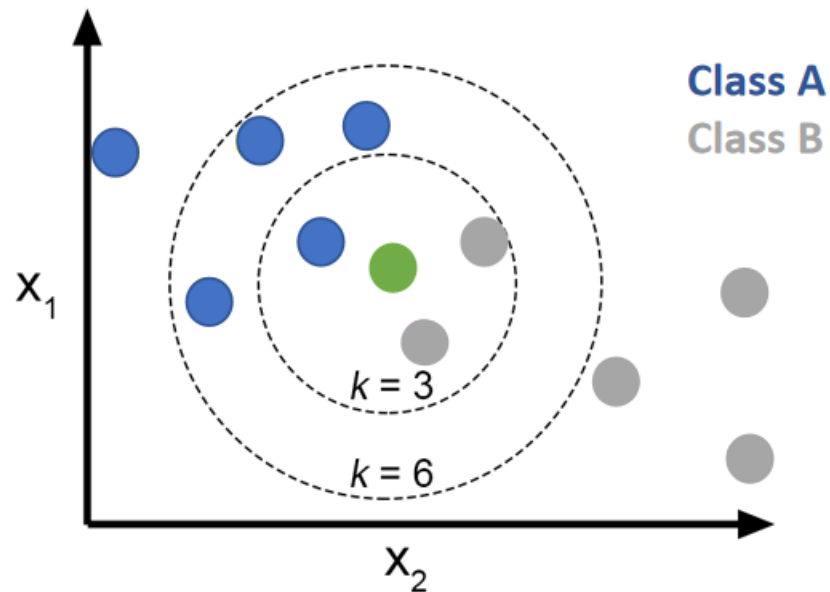
- To find the most suitable algorithm, we investigate four popular machine learning algorithms and make a **comparison** among them.



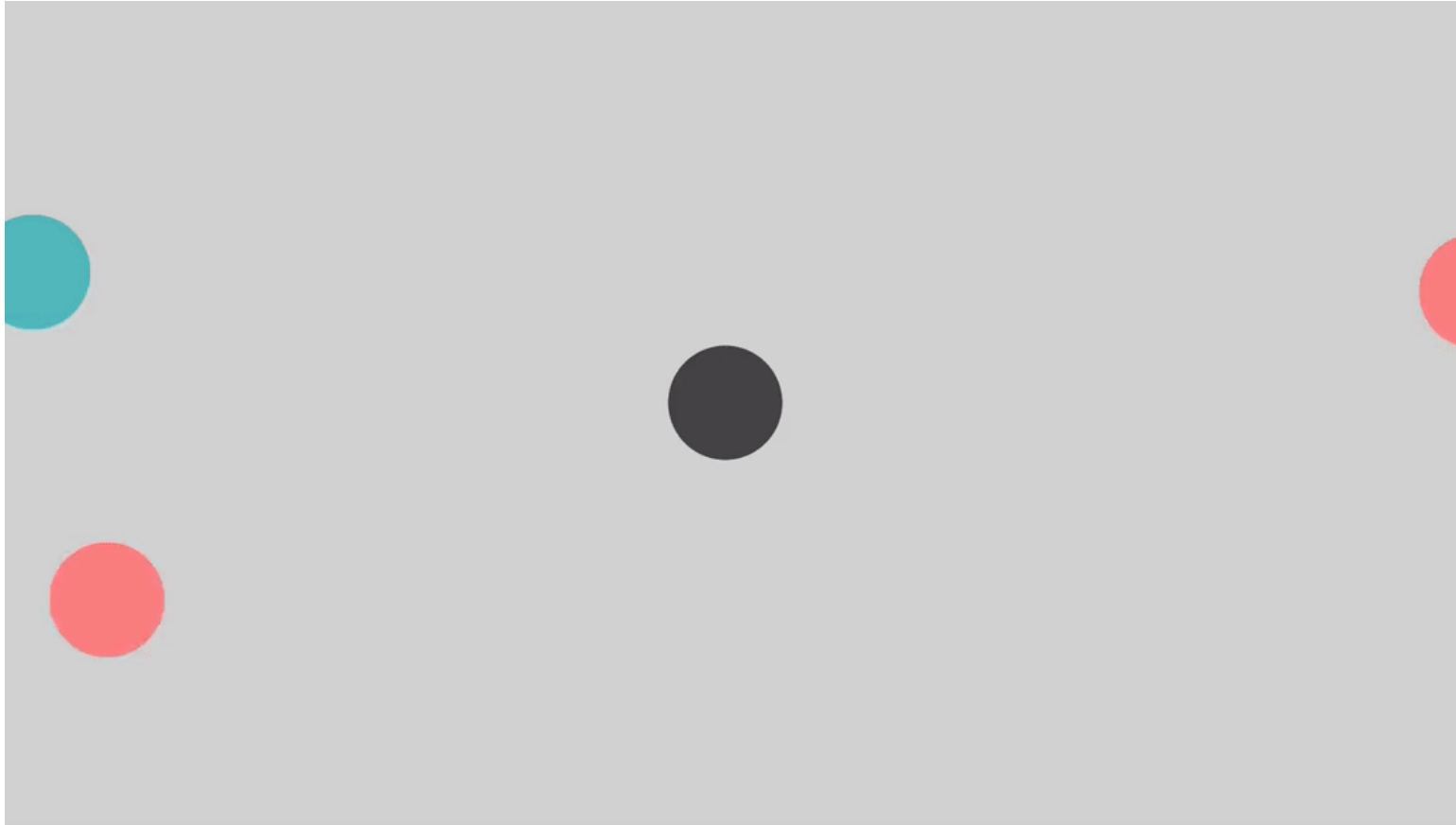
- K Nearest Neighbor (KNN)
- Back Propagate Neural Network (BPNN)
- Support Vector Classification (SVC)
- Naïve Bayes (NB)

K-nearest neighbors method

- One of the simplest **machine learning algorithms**
- Used for **classification**
- Based on the **similarity (distance)**



K-nearest neighbors method



(kNN in a classification with $k=3$.) An object is classified by a plurality vote of its 3 neighbors, with the object being assigned to the class most common among its 3 nearest neighbors.

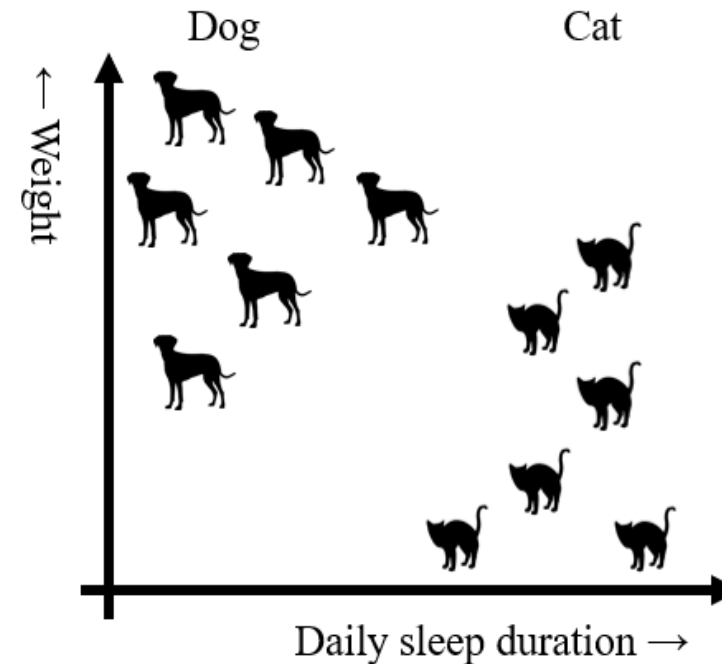
An example

■ Step 1: Choose **factors**

Weight, daily sleep duration.....

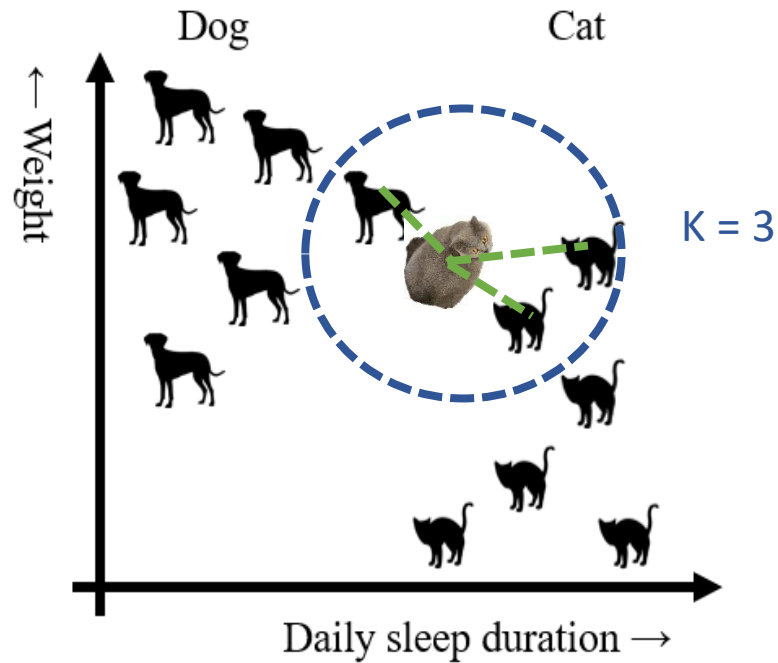
■ Step 2: Prepare **training data**

Number	Type	Weight (kg)	Daily Sleep Duration (hour)
1	dog	17	8
2	cat	6	16
3	dog	10	10
.....			

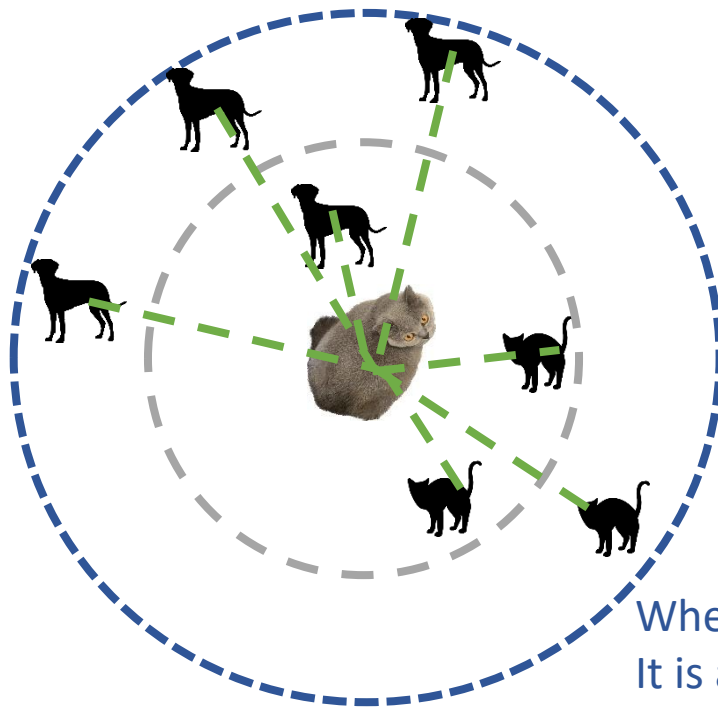


An example

- Step 3: Input **testing data**
- Step 4: Choose the value of **K** and make predictions



The value of K



- A **user-defined** constant
- Represents the **number of the nearest neighbors**
- Affects the **accuracy** of prediction

KNN implementation

Now, let's try to use KNN to predict students' grade in ENGG2020.

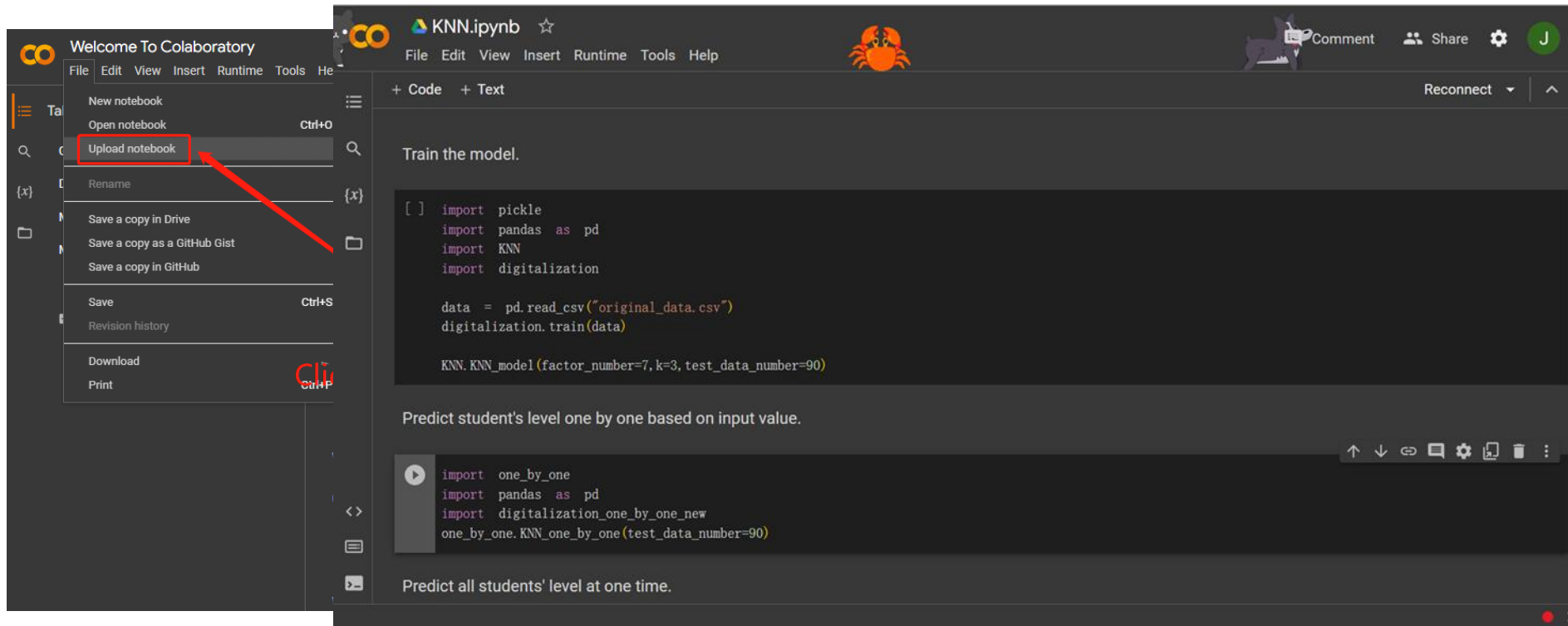
- Step 1: Choose **factors**
- Step 2: Prepare **training data**
- Step 3: Input **test data**
- Step 4: Choose different values of **K** and make predictions

Now it is your turn!



Preparations

- Upload the “KNN.ipynb” from the file “Main_code”.



类型	大小
PYNB 文件	15 KB
PYNB 文件	137 KB
PYNB 文件	84 KB
PYNB 文件	82 KB

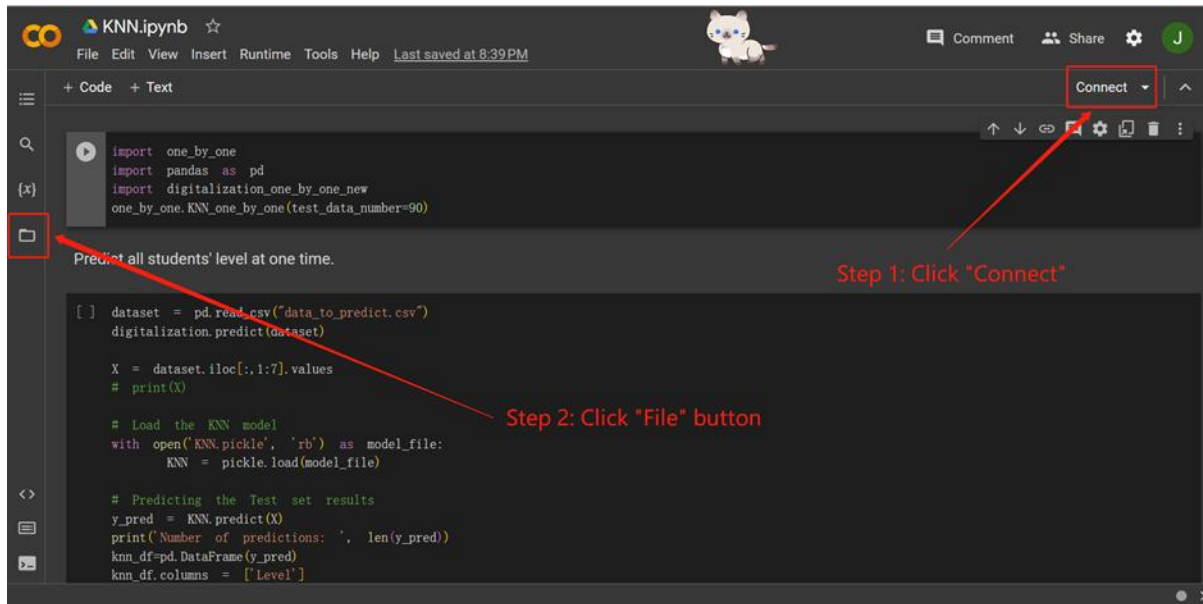
Upload the program

Upload complete

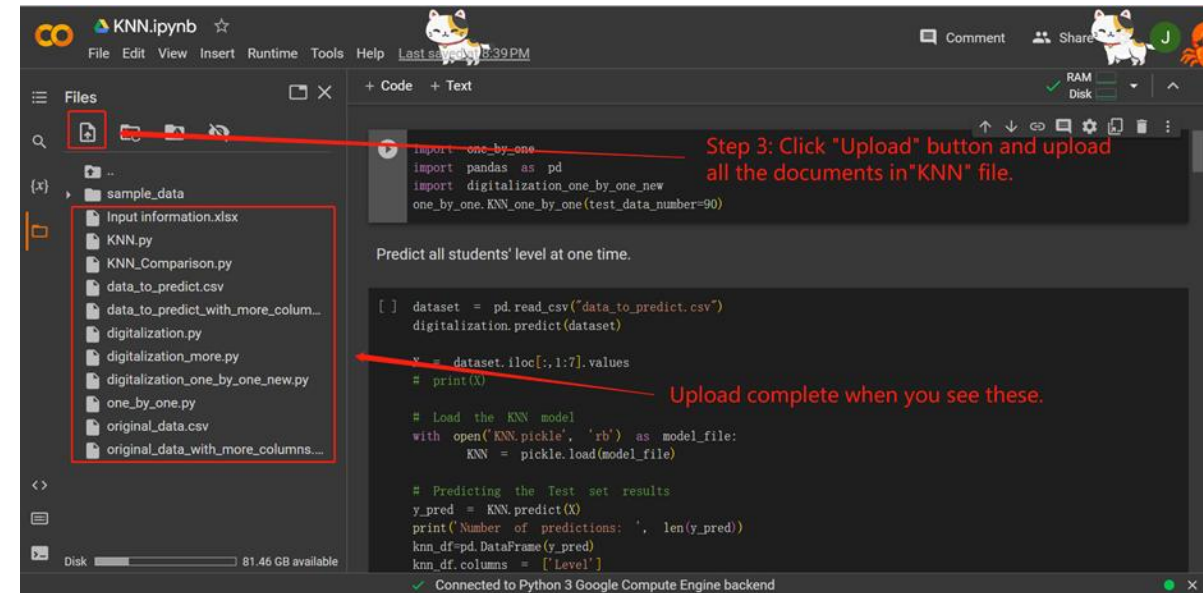
Select “KNN.ipynb”

Preparations

- **Connect** the program to online processor.
- **Upload** all the documents from “KNN” file to the notebook.



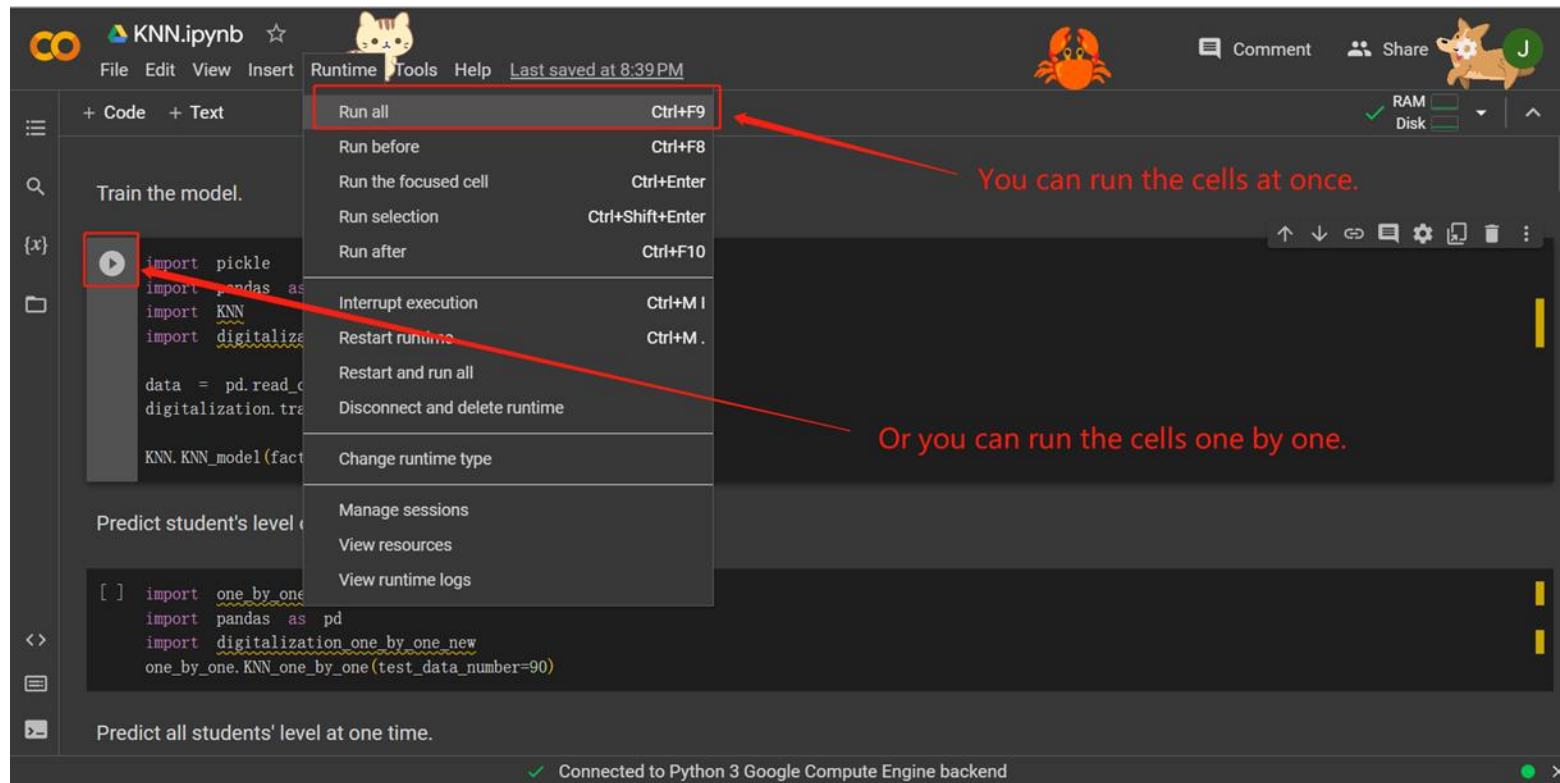
Connect the notebook



Upload our files

Run the program

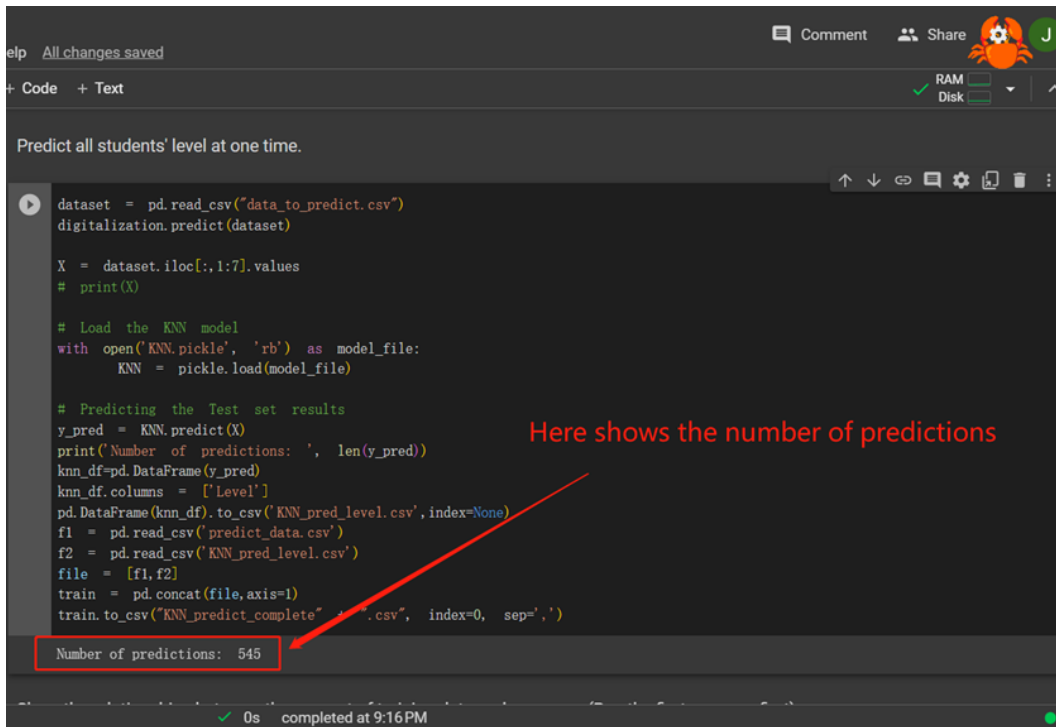
- Click “Run” button and **run the cells one by one** or you can run them all in one click. (One by one is recommended)



Run the KNN program

Check the results

- Click “Run” button and run the cells one by one or you can run them all in one click. (One by one is recommended)



```
dataset = pd.read_csv("data_to_predict.csv")
digitalization.predict(dataset)

X = dataset.iloc[:, 1:7].values
# print(X)

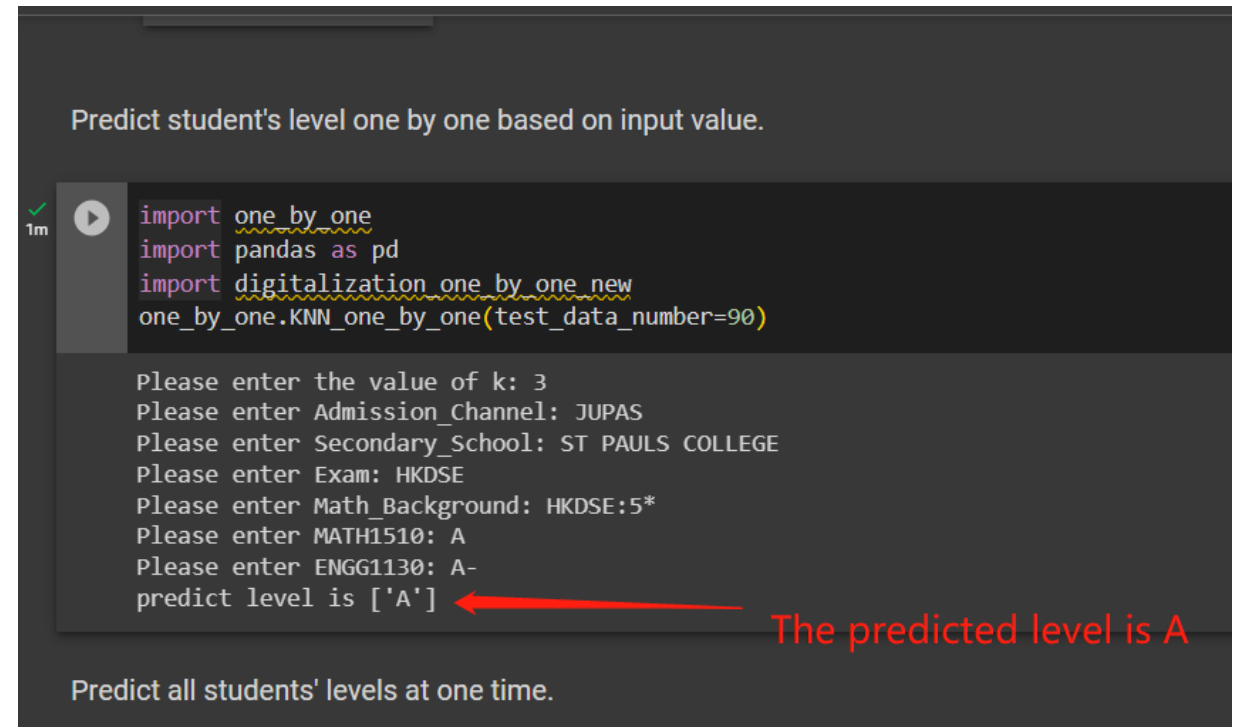
# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    KNN = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
knn_df=pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1, f2]
train = pd.concat(file,axis=1)
train.to_csv("KNN_predict_complete" + ".csv", index=0, sep=',')
```

Number of predictions: 545

0s completed at 9:16 PM

Output of first cell



```
import one_by_one
import pandas as pd
import digitalization one_by_one new
one_by_one.KNN_one_by_one(test_data_number=90)
```

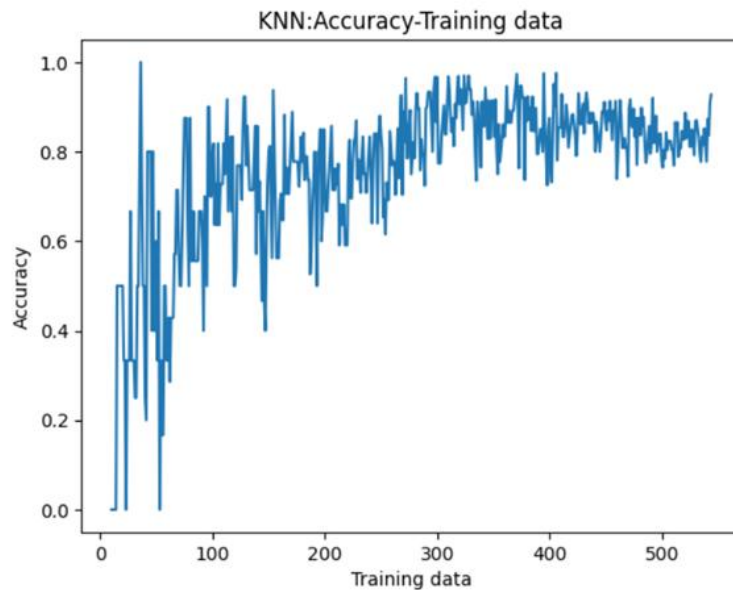
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
Please enter ENGG1130: A-
predict level is ['A']

Predict all students' levels at one time.

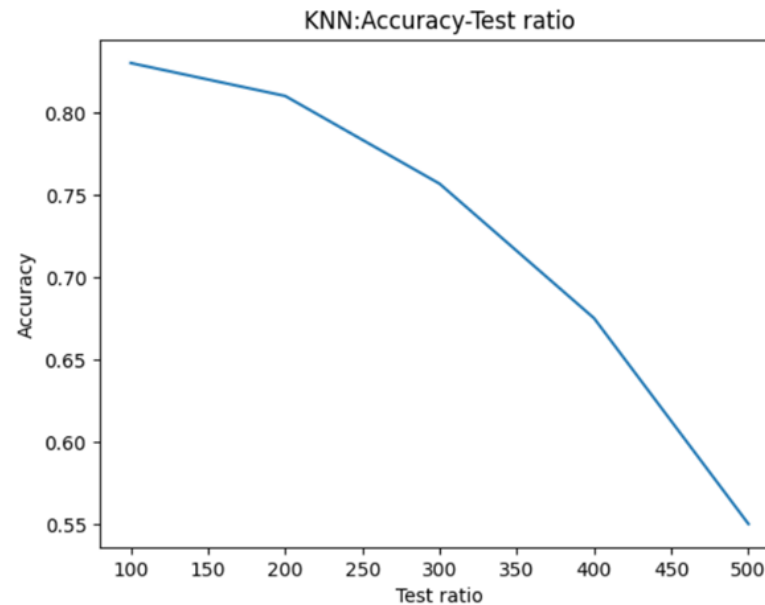
Output of second cell

Check the results

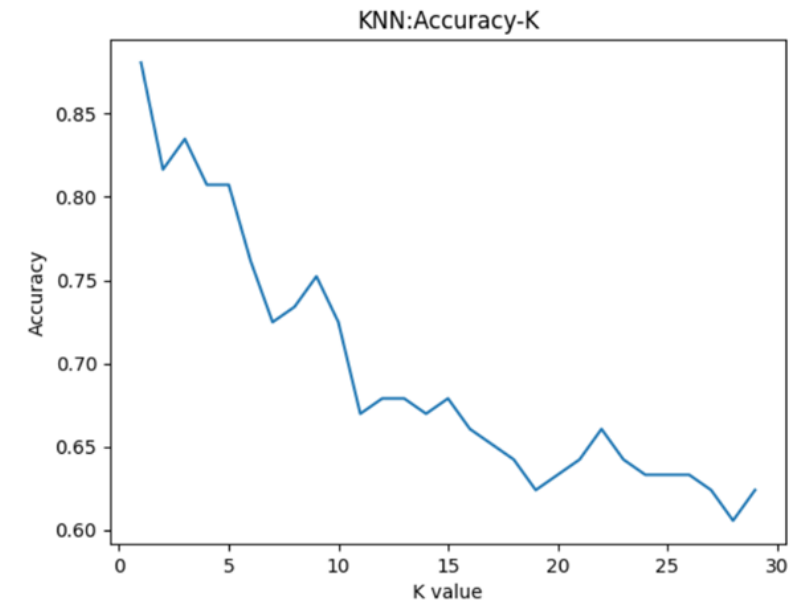
■ Click “Run” button and [run the cells one by one](#) or you can run them all in one click.



Output of third cell



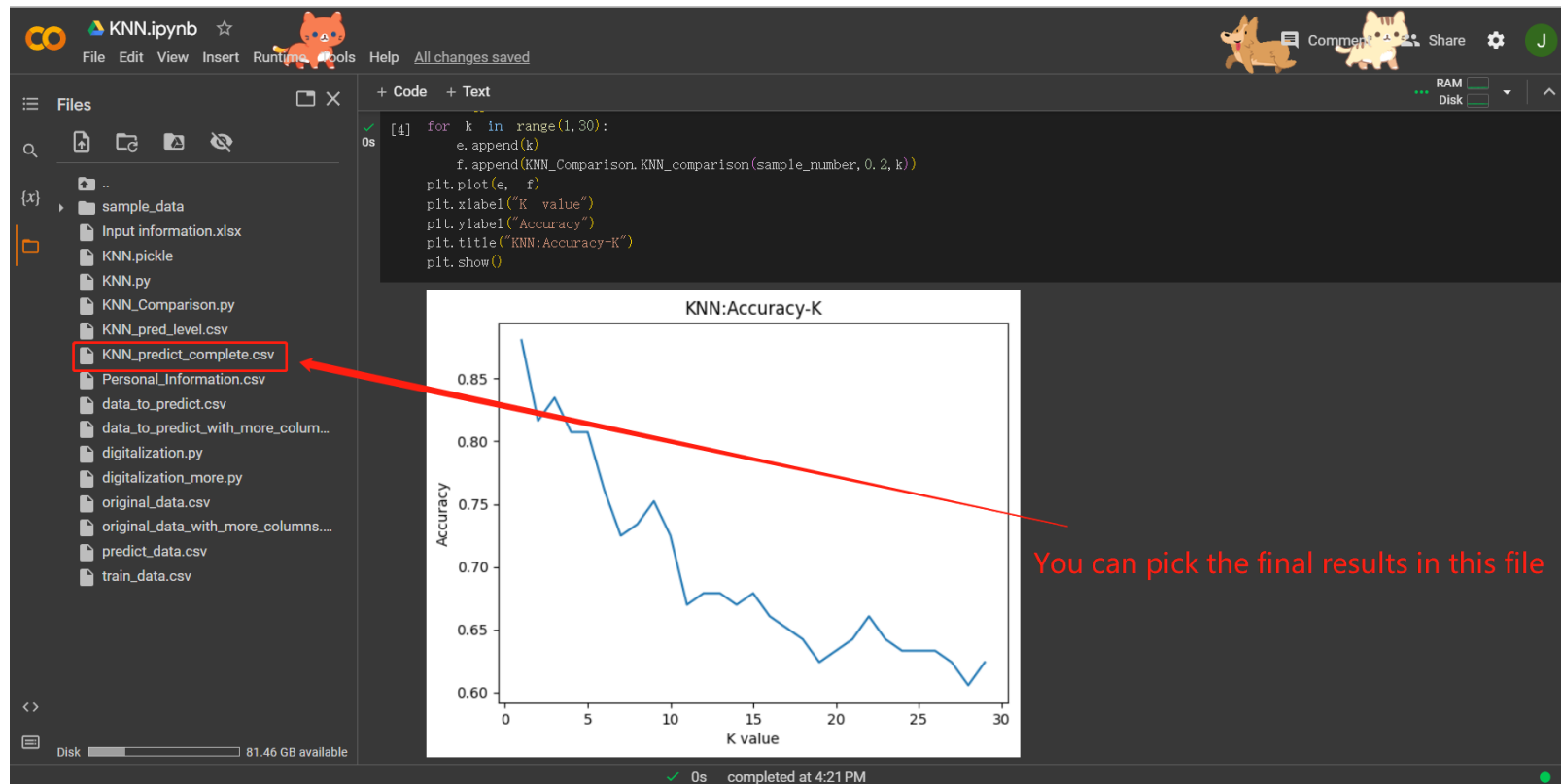
Output of fourth cell



Output of fifth cell

Check the outputs

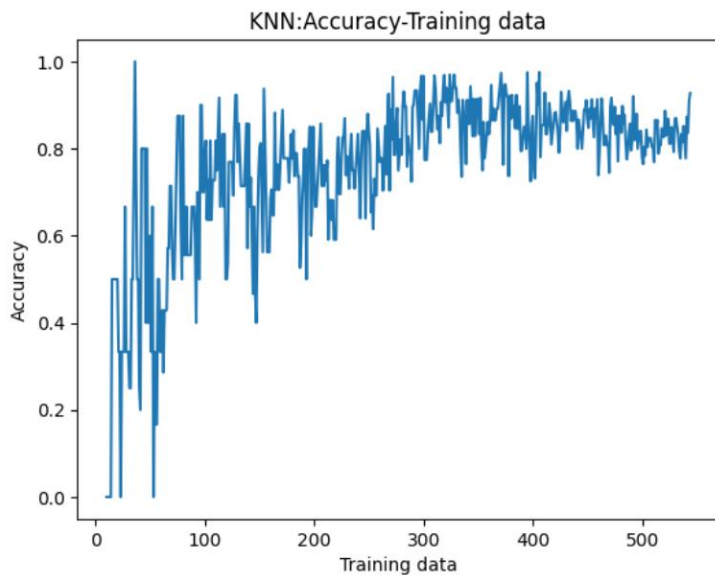
- The final output can be found on the left, named “KNN_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.



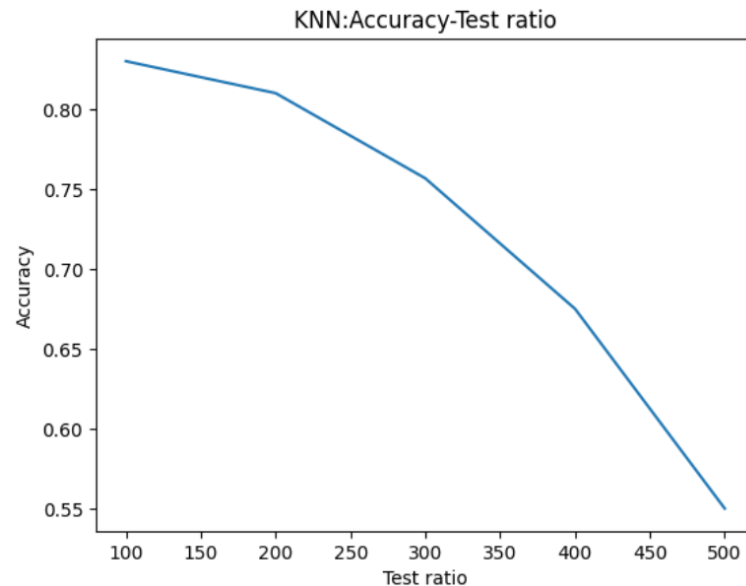
Outcome file

Select the optimal parameters

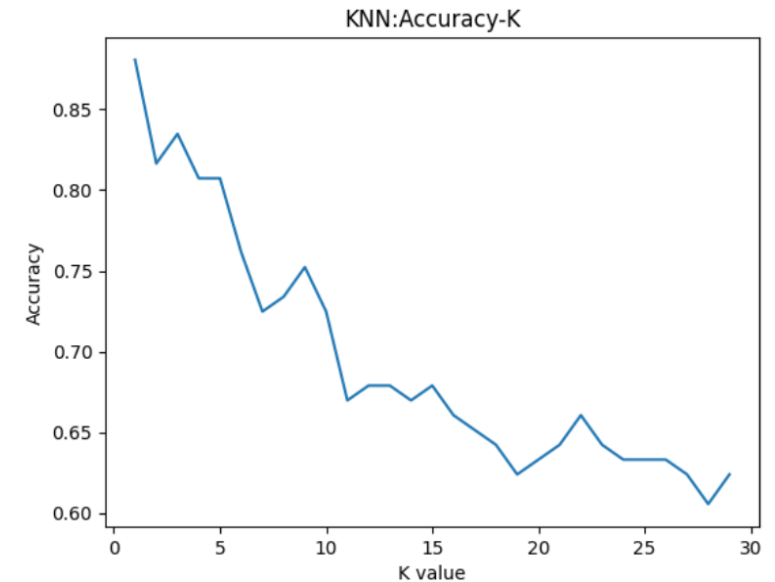
- By comparing the prediction data and the target, we can calculate the **accuracy** of the trained model.
- For each algorithm, we can also find out in which set of **parameters** the model can reach the highest accuracy.



Generally, more training data represents higher accuracy



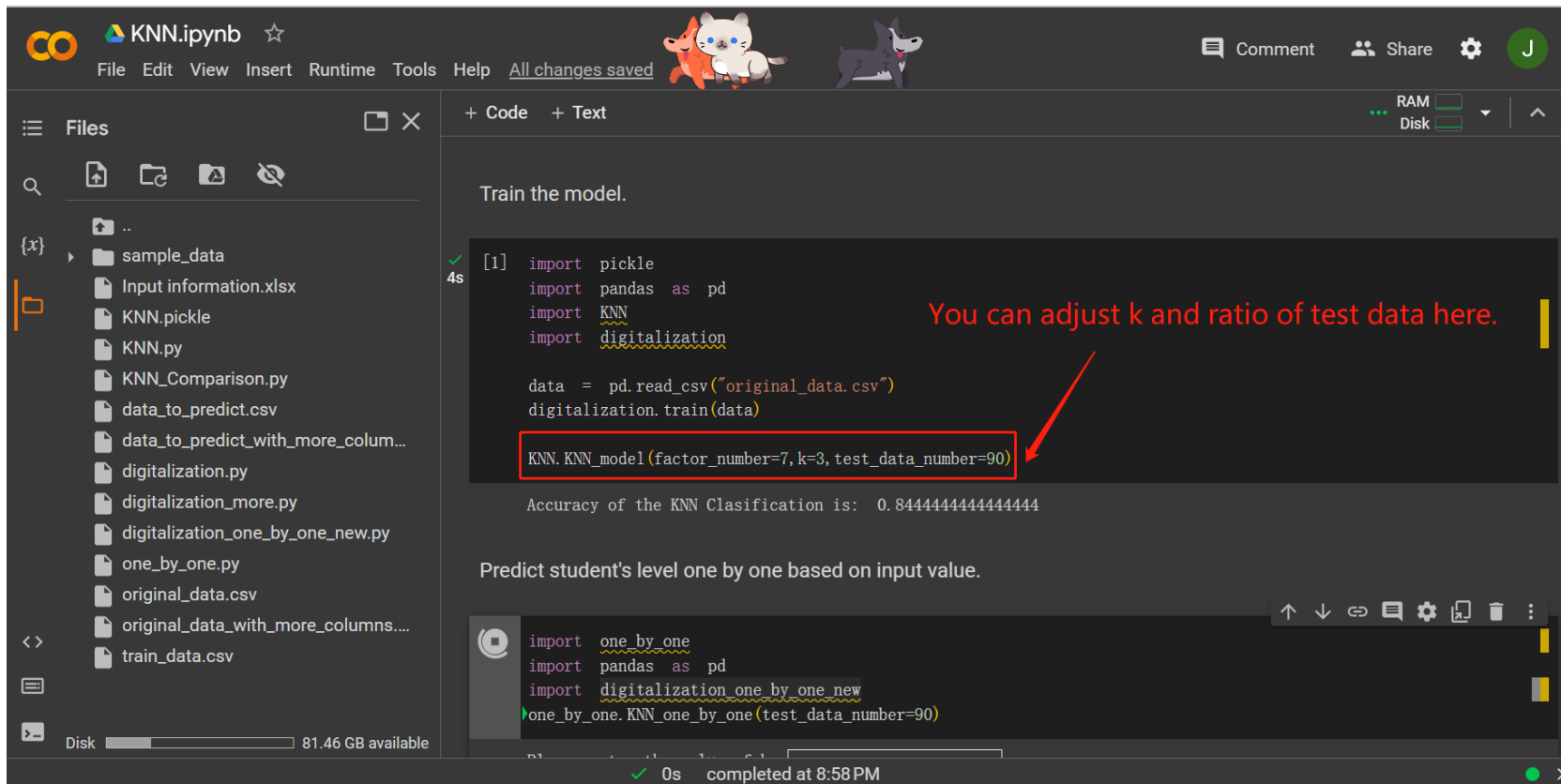
Generally, less test data represents higher accuracy



K=3 or K=1 provides a desired model

Select the optimal parameters

- After comparison, you can adjust the K value and ratio of test data like in the picture below.



```
File Edit View Insert Runtime Tools Help All changes saved
KNN.ipynb
Files
sample_data
Input information.xlsx
KNN.pickle
KNN.py
KNN_Comparison.py
data_to_predict.csv
data_to_predict_with_more_column...
digitalization.py
digitalization_more.py
digitalization_one_by_one_new.py
one_by_one.py
original_data.csv
original_data_with_more_columns...
train_data.csv
+ Code + Text
Train the model.
[1] import pickle
import pandas as pd
import KNN
import digitalization

data = pd.read_csv("original_data.csv")
digitalization.train(data)

KNN.KNN_model(factor_number=7, k=3, test_data_number=90)

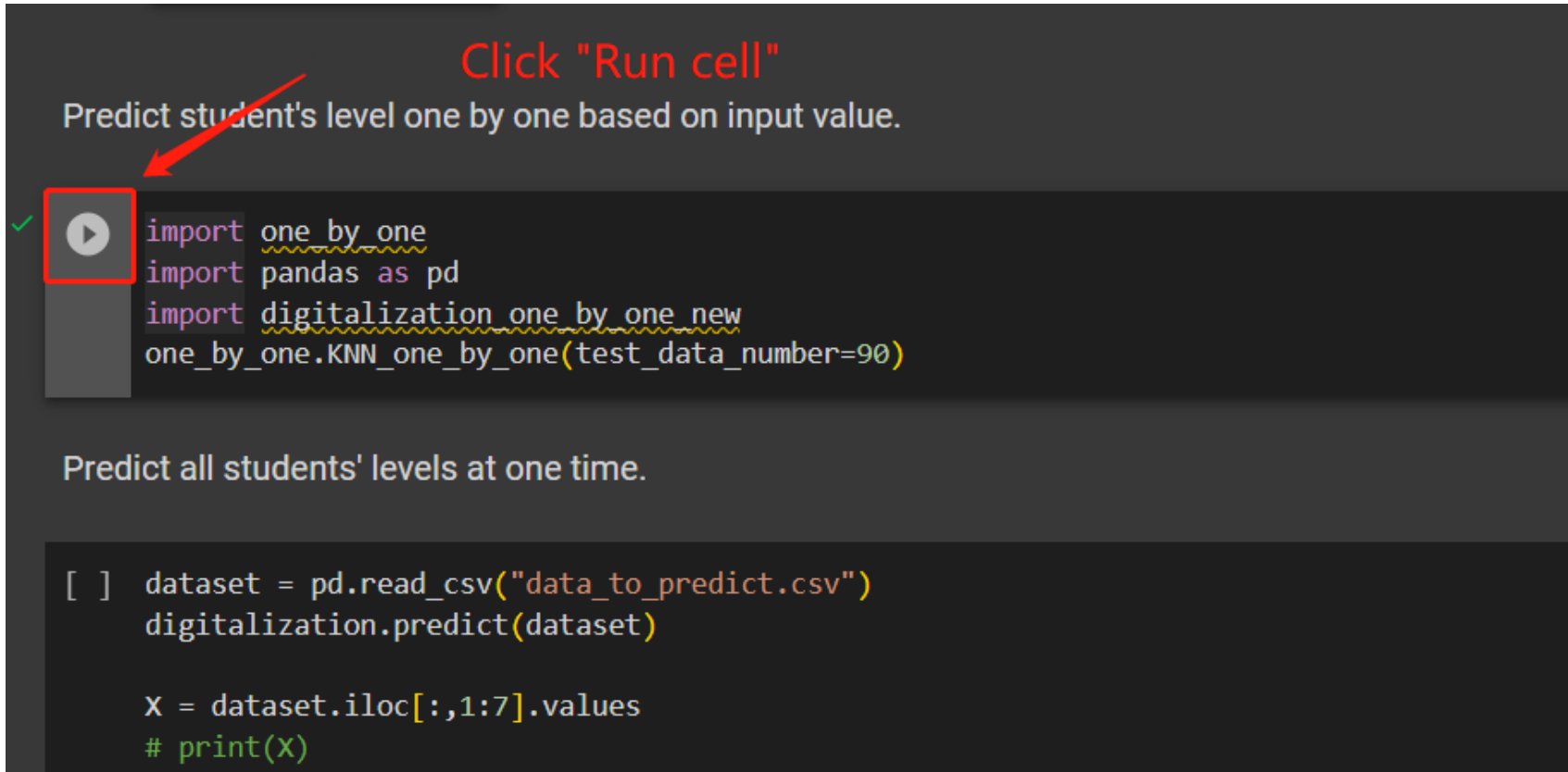
Accuracy of the KNN Clasification is: 0.8444444444444444

Predict student's level one by one based on input value.
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
0s completed at 8:58 PM
```

You can adjust the parameters here based on the comparison

Predict one by one

- Run the following code



The screenshot shows a Jupyter Notebook interface with a dark background. At the top, there is a text prompt: "Predict student's level one by one based on input value." Below this, there is a code cell. To the left of the code cell is a green checkmark and a play button icon (a circle with a right-pointing triangle) inside a red square. A red arrow points from the text "Click 'Run cell'" to the play button icon. The code in the cell is as follows:

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Below the code cell, there is another text prompt: "Predict all students' levels at one time." Below this, there is another code cell with the following code:

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)

X = dataset.iloc[:,1:7].values
# print(X)
```

Predict one by one

- Select a value of K

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k:

Predict all students' levels at one time.

Choose a value of k

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)
```

Predict one by one

- Enter **Admission_Channel**: JUPAS/Mainland Scholarship(Tuition)/Mainland
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3

Please enter Admission_Channel:

Predict all students' levels at one time.

Choose an Admission_Channel

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)

X = dataset.iloc[:,1:7].values
```

Predict one by one

- Enter **Secondary_School**: For example: LA SALLE COLLEGE/ ST PAULS COLLEGE/SHA TIN METHODIST COLLEGE/Others
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School:

Choose a Secondary_School

Predict all students' levels at one time.

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
digitalization_predict(dataset)
```

Predict one by one

- Enter **Exam type:** HKDSE/Mainland Gaokao (JEE)/IB/GCE-AL...
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary School: ST PAULS COLLEGE
Please enter Exam:

Predict all students' levels at one time. Choose an exam

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)
```

Predict one by one

- **Math_Background:** HKDSE:5**/HKDSE:5/Mainland Gaokao (JEE):>140/ Mainland Gaokao (JEE):130-139...
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE

Please enter Math_Background:

Choose a Math_Background

Predict all students' levels at one time.

Predict one by one

- **MATH1510:** A/A-/B+/B/B-/C+/C/C-/D+/D
- And then click “Enter”

```
Predict student's level one by one based on input value.
```

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

```
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
```

Choose a grade of MATH1510

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
digitalization_predict(dataset)
```

Predict one by one

- **ENGG1130:** A/A-/B+/B/B-/C+/C/C-/D+/D/F
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: 5*
Please enter MATH1510: A-

Please enter ENGG1130:

Choose a grade of ENGG1130

Predict all students' levels at one time.

Predict one by one

- Then we can get the predicted level of the student.

```
Predict student's level one by one based on input value.
```

```
✓ 1m ▶ import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

```
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
Please enter ENGG1130: A-
predict level is ['A']
```

The predicted level is A

```
Predict all students' levels at one time.
```

Want to try your own data?

■ Follow the steps below:

1. First, open the “**Input information.xlsx**” from the “KNN” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.

	A	B	C	D	E	F	G
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE):>140	A	A
3	2	Non-JUPAS (Local)	YUEN WAN PUB HO CHUEN YIU MEMORIAL CO	HKDSE	HKDSE:5*	B+	C
4							
5							
6							
7			SKH LEUNG KWAI YEE SECONDARY SCHOOL				
8			SKH BISHOP BAKER SECONDARY SCHOOL				
9			SKH BISHOP MOK SAU TSENG SEC SCH				
10			SKH TANG SHIU KIN SECONDARY SCHOOL				
11			SKH CHAN YOUNG SECONDARY SCHOOL				
12			ST STEPHEN'S GIRLS' COLLEGE				
13			ST BONAVENTURE COLLEGE & HIGH SCHOOL				
14			IMMACULATE HEART OF MARY COLLEGE				
15			OUR LADY OF THE ROSARY COLLEGE				
16			ST ROSE OF LIMA'S COLLEGE				
17			ST FRANCIS XAVIER'S COLLEGE				
18			ST JOSEPH'S COLLEGE				
19							
20							

Input your information

Want to try your own data?

2. **Upload** the “Input information.xlsx” to program like before and run the **fifth cell**.

3. Then pick up the “**Personal_KNN_predict_complete.csv**” file in the left to check the prediction of your grade!

```
import digitalization
df = pd.read_excel('Input information.xlsx')
df1 = df.drop(['School_Information'], axis=1)
pd.DataFrame(df1).to_csv('Personal_Information.csv', index=None)
dataset = pd.read_csv('Personal_Information.csv')
digitalization.predict(dataset)
student_number = int(input('Please input the number of students: '))
X = dataset.iloc[:student_number, 1:7].values
# print(X)

# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    KNN = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
print('Predict Level: ', y_pred)
knn_df = pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1, f2]
train = pd.concat(file, axis=1)
train.to_csv('Personal_KNN_predict_complete' + '.csv', index=0, sep=',')
```

Please input the number of students: 2
Number of predictions: 2
Predict Level: ['A' 'C']

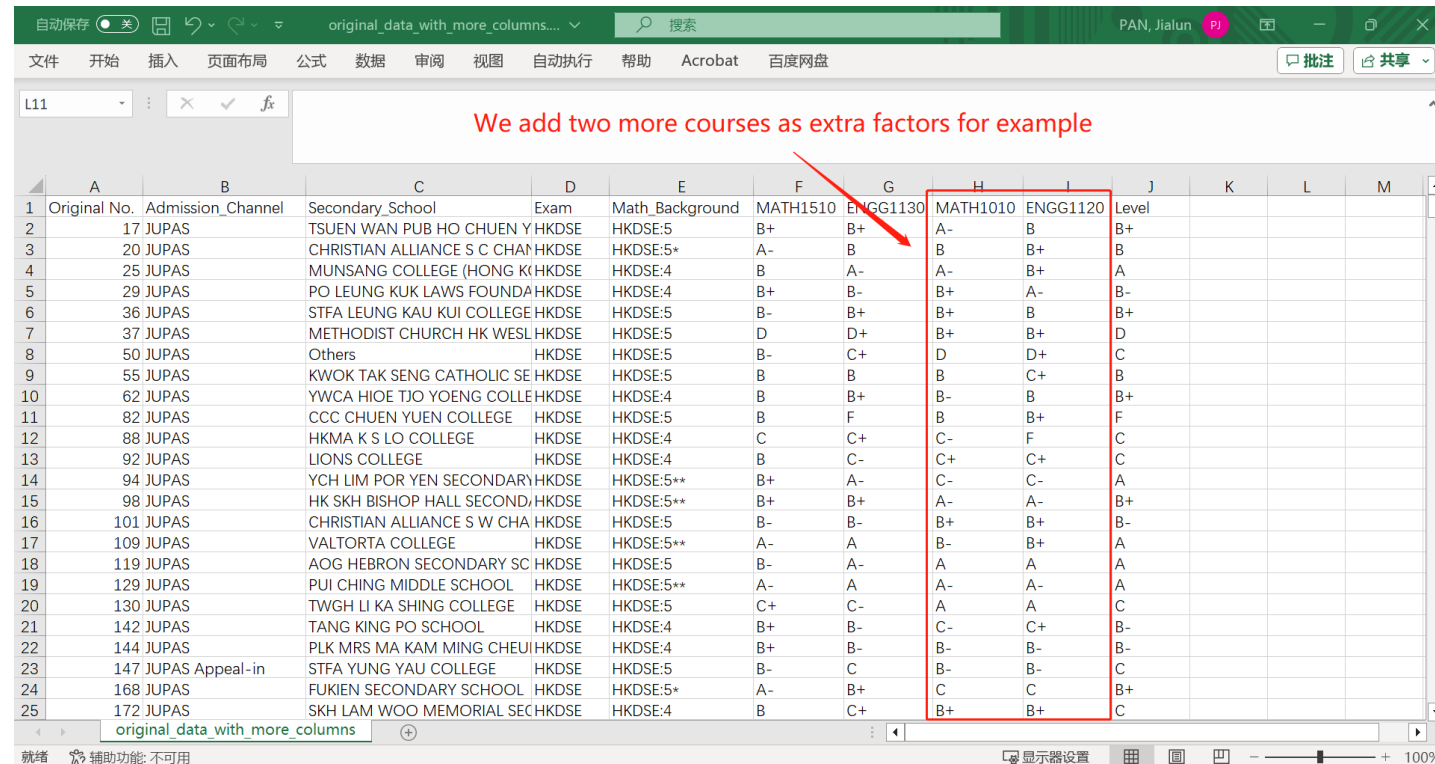
Run the seventh cell.

Two ways to see the predictions

Want to add extra factors?

■ Follow the steps below:

1. We can take a look at “original_data_with_more_columns.csv” in “KNN” file.



We add two more courses as extra factors for example

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	Level			
1	17	JUPAS	TSUEN WAN PUB HO CHUEN Y HKDSE	HKDSE:5	B+	B+		A-	B	B+			
2	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN HKDSE	HKDSE:5*	A-	B		B	B+	B			
3	25	JUPAS	MUNSANG COLLEGE (HONG KONG) HKDSE	HKDSE:4	B	A-		A-	B+	A			
4	29	JUPAS	PO LEUNG KUK LAWS FOUNDATION HKDSE	HKDSE:4	B+	B-		B+	A-	B-			
5	36	JUPAS	STFA LEUNG KAU KUI COLLEGE HKDSE	HKDSE:5	B-	B+		B+	B	B+			
6	37	JUPAS	METHODIST CHURCH HK WESLEYAN HKDSE	HKDSE:5	D	D+		B+	B+	D			
7	50	JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C			
8	55	JUPAS	KWOK TAK SENG CATHOLIC SEHKDSE	HKDSE:5	B	B		B	C+	B			
9	62	JUPAS	YWCA HIOE TJO YOENG COLLEGE HKDSE	HKDSE:4	B	B+		B-	B	B+			
10	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F			
11	88	JUPAS	HKMA K S LO COLLEGE	HKDSE:4	C	C+		C-	F	C			
12	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C			
13	94	JUPAS	YCH LIM POR YEN SECONDARY HKDSE	HKDSE:5**	B+	A-		C-	C-	A			
14	98	JUPAS	HK SKH BISHOP HALL SECONDARY HKDSE	HKDSE:5**	B+	B+		A-	A-	B+			
15	101	JUPAS	CHRISTIAN ALLIANCE S W CHAN HKDSE	HKDSE:5	B-	B-		B+	B+	B-			
16	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A			
17	119	JUPAS	AOG HEBRON SECONDARY SCHOOL HKDSE	HKDSE:5	B-	A-		A	A	A			
18	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A			
19	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C			
20	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-			
21	144	JUPAS	PLK MRS MA KAM MING CHEUNG HKDSE	HKDSE:4	B+	B-		B-	B-	B-			
22	147	JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C			
23	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE:5*	A-	B+		C	C	B+			
24	172	JUPAS	SKH LAM WOO MEMORIAL SECONDARY HKDSE	HKDSE:4	B	C+		B+	B+	C			

You can replace the content as you like

Want to add extra factors?

2. After training the model with the dataset in previous slide.
3. You can make a csv document to **contain the students' information** you want to predict their level.

After running the sixth cell, the level of these students will be added here

	A	B	C	D	E	F	G	H	I	J	K
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120		
2	627	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-		
3	882	JUPAS	CHRISTIAN ALLIANCE CHENG WING GEE C	HKDSE	HKDSE:4	D	F	B-	B+		
4	801	JUPAS	MA ON SHAN TSUNG TSIN SECONDARY S	HKDSE	HKDSE:5*	B-	B	C	C+		
5	1087	Mainland Fee-paying	Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+		
6	224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED	HKDSE	HKDSE:5*	B-	F	B-	B-		
7	240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+		
8	911	JUPAS	Others	HKDSE	HKDSE:5**	A	A	B-	B+		
9	917	JUPAS	DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+		
10	624	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-1A-	A-	A	B	A		
11	724	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+		
12	546	JUPAS	KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+		
13	410	Mainland	Others	Mainland	(Mainland Gaokao (JEE):130-1B+	B+	B	A-	B		
14	492	JUPAS	Others	HKDSE	HKDSE:5	B-	A-	B	B-		
15	986	JUPAS	Others	HKDSE	HKDSE:4	C+	B+	A-	B+		
16	1046	JUPAS	HELEN LIANG MEMORIAL SEC SCH (SHATIN	HKDSE	HKDSE:5	A-	B+	B	B		
17	272	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-		
18	803	JUPAS	Others	HKDSE	HKDSE:4	B-	B+	B	B-		
19	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B		
20	881	JUPAS	YAN OI TONG TIN KA PING SECONDARY S	HKDSE	HKDSE:4	B-	C+	B+	B		
21	932	JUPAS	SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-		
22	120	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-1A	A	A	A-	A		
23	732	JUPAS	HKSYC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+		
24	474	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-		
25	776	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-		

The level of students will be added automatically

Want to add extra factors?

4. Then give the file to our model.

The screenshot shows a Jupyter Notebook titled "KNN.ipynb". The left sidebar displays a file explorer with the following files: `sample_data`, `Input information.xlsx`, `KNN.pickle`, `KNN.py`, `KNN_Comparison.py`, `KNN_pred_level.csv`, `KNN_predict_complete.csv`, `More_KNN_predict_complete.csv` (highlighted with a yellow box), `Personal_Information.csv`, `Personal_KNN_predict_complete.csv`, `data_to_predict.csv`, `data_to_predict_with_more_columns.csv` (highlighted with a red box), `digitalization.py`, `digitalization_more.py`, `original_data.csv`, `original_data_with_more_columns.csv` (highlighted with a red box), `predict_data.csv`, and `train_data.csv`.

The main code cell contains the following Python code:

```
[6] import pickle
import pandas as pd
import KNN
import digitalization_more

data = pd.read_csv("original_data_with_more_columns.csv")
digitalization_more.train(data)

KNN.KNN_model(8)

dataset = pd.read_csv("data_to_predict_with_more_columns.csv")
digitalization_more.predict(dataset)

X = dataset.iloc[:, 1:8].values
# print(X)

# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    knn = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
knn_df=pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv('More_KNN_predict_complete' + ".csv", index=0, sep=',')
```

Annotations and steps:

- Step 1:** Upload two csv files to the program. (Red arrow pointing to `data_to_predict.csv` and `data_to_predict_with_more_columns.csv` in the file explorer).
- Step 2:** You can check the accuracy of the predictions. (Blue arrow pointing to the output of the code cell).
- Step 3:** You can pick the final results in this csv file after running the cell. (Yellow arrow pointing to `More_KNN_predict_complete.csv` in the file explorer).

The output of the code cell shows:

```
Accuracy of the KNN Clasification is: 0.6330275229357798
Number of predictions: 545
```

The status bar at the bottom indicates "0s completed at 4:32 PM".

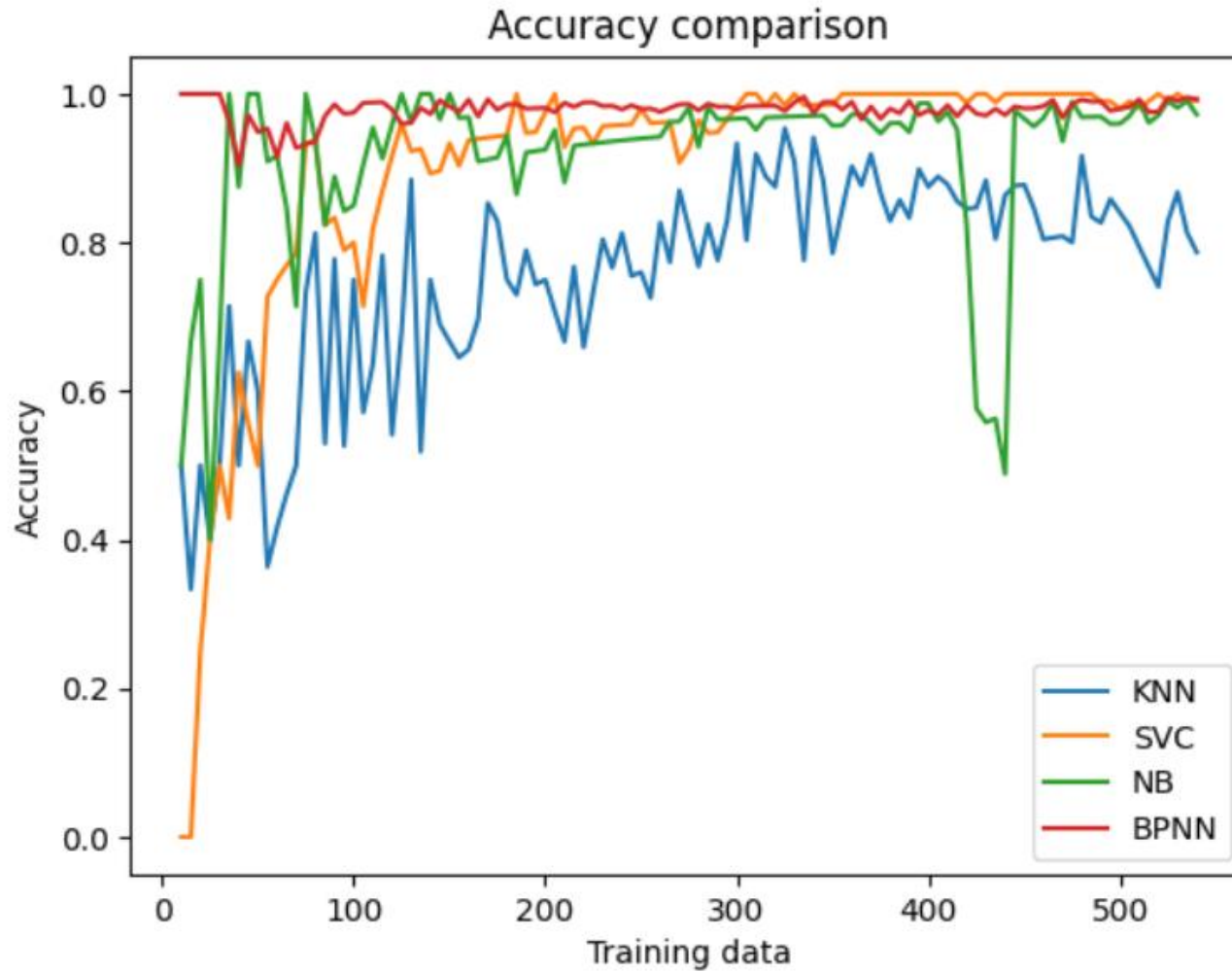
Run the eighth cell and get your results



Methods comparison

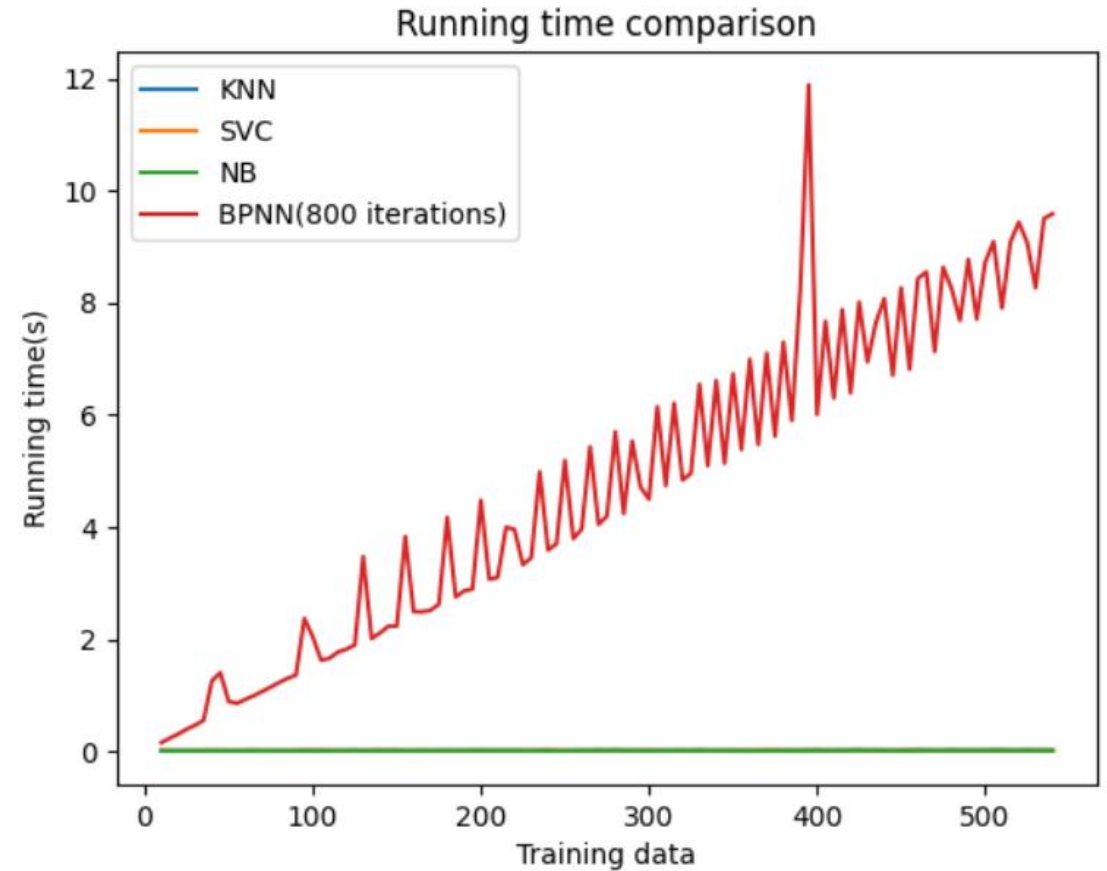
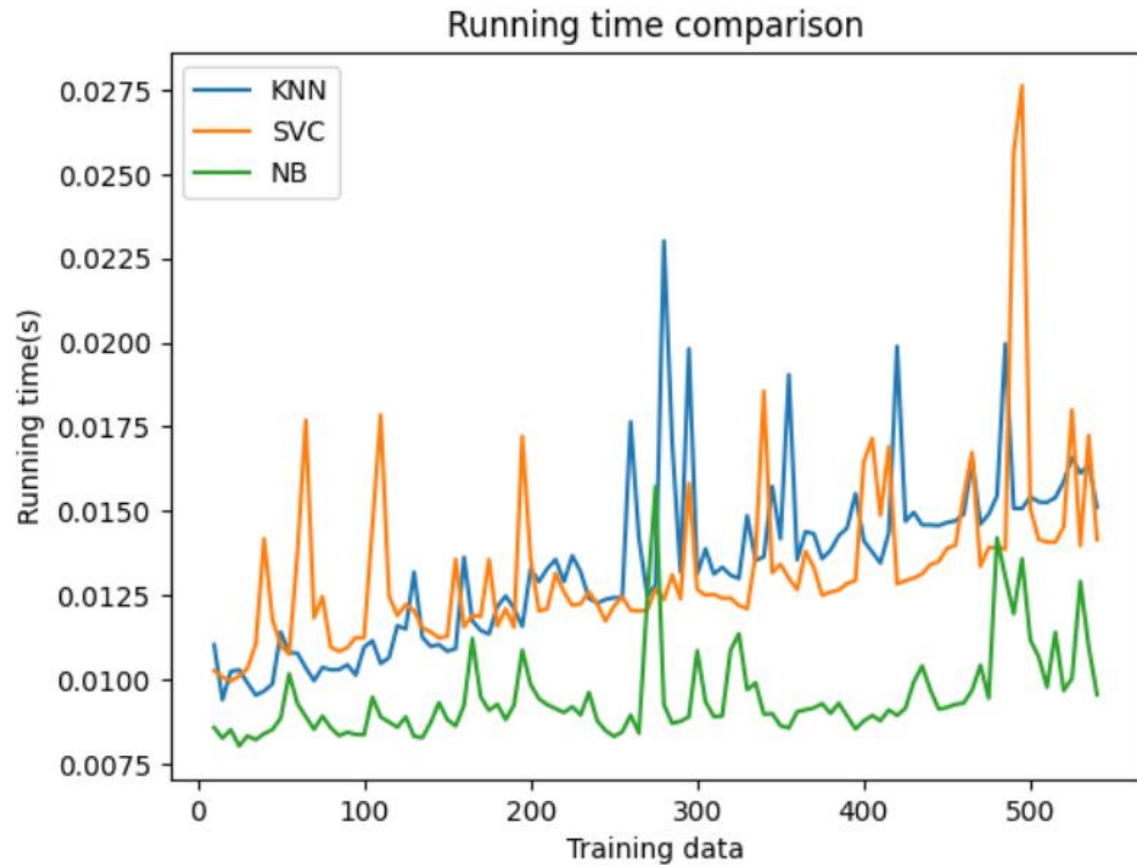
Select the best methods for
learner level prediction

Comparison in accuracy



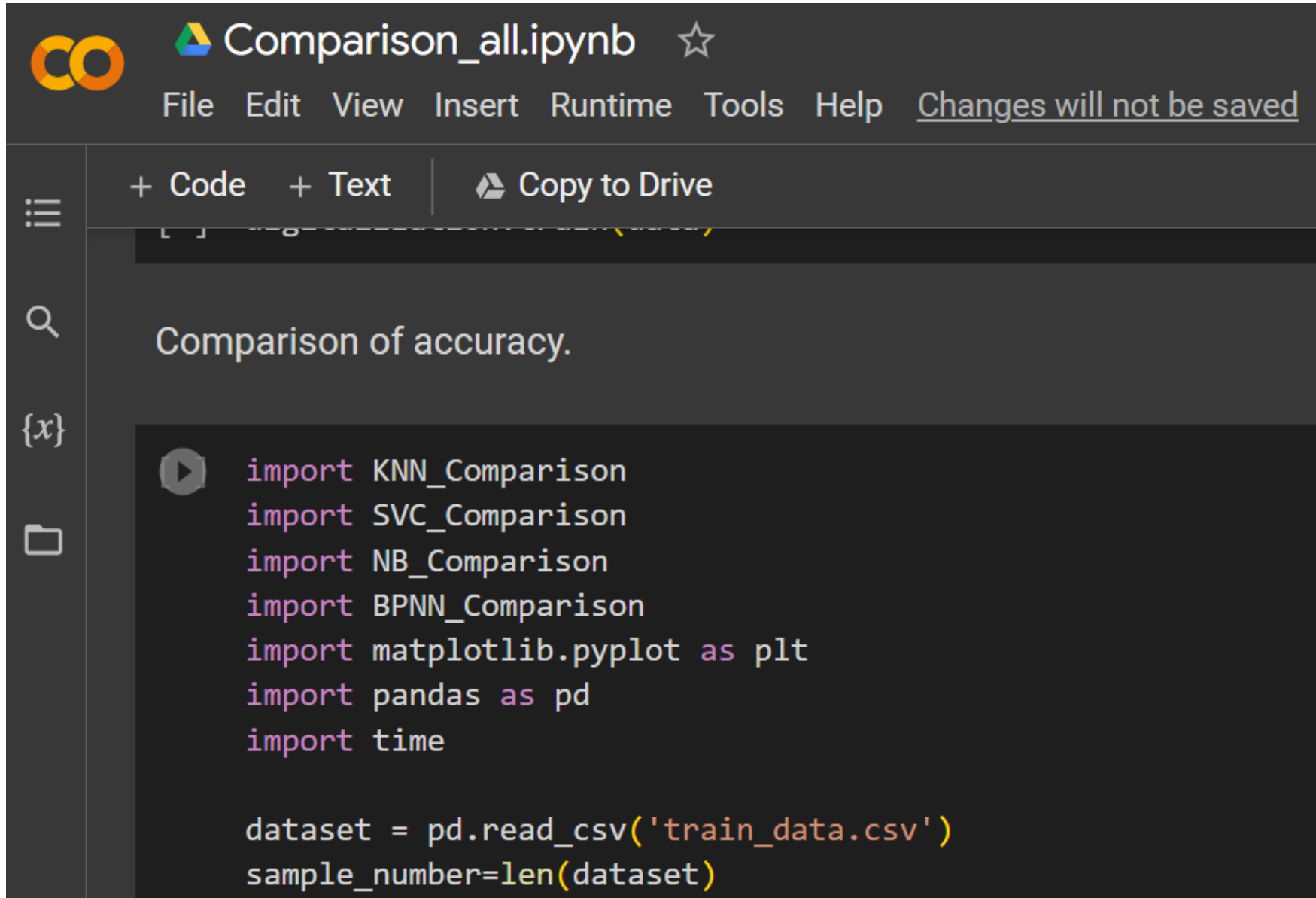
Comparison of four algorithms in accuracy

Comparison in accuracy



Comparison of four algorithms in computational time

Code for comparison

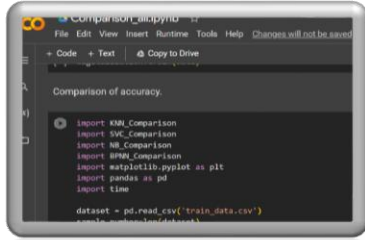


The image shows a Jupyter Notebook interface with a dark theme. The top bar is green with the text "Code for comparison". The notebook title is "Comparison_all.ipynb" with a star icon. The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", "Help", and a warning "Changes will not be saved". The left sidebar has icons for a menu, search, variables, and a file explorer. The main area has a tab for "Code" and a "Copy to Drive" button. The text "Comparison of accuracy." is displayed above a code cell. The code cell contains the following Python code:

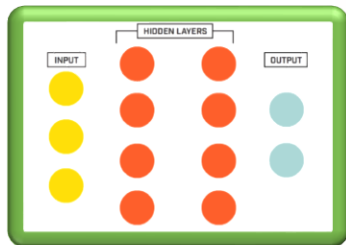
```
import KNN_Comparison
import SVC_Comparison
import NB_Comparison
import BPNN_Comparison
import matplotlib.pyplot as plt
import pandas as pd
import time

dataset = pd.read_csv('train_data.csv')
sample_number=len(dataset)
```

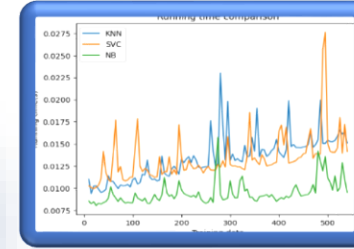
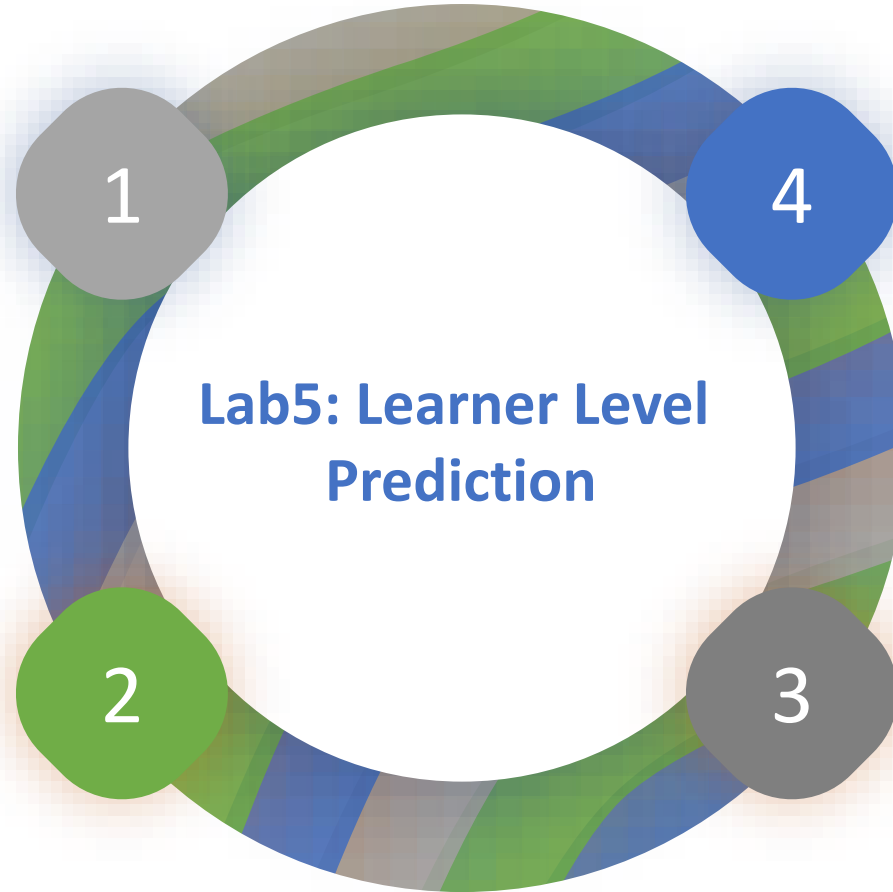
Takeaways



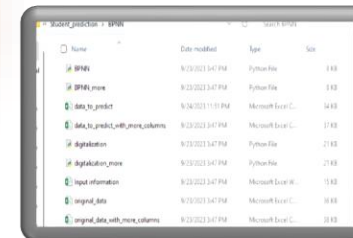
Operation
on Colab



4 methods
introduction

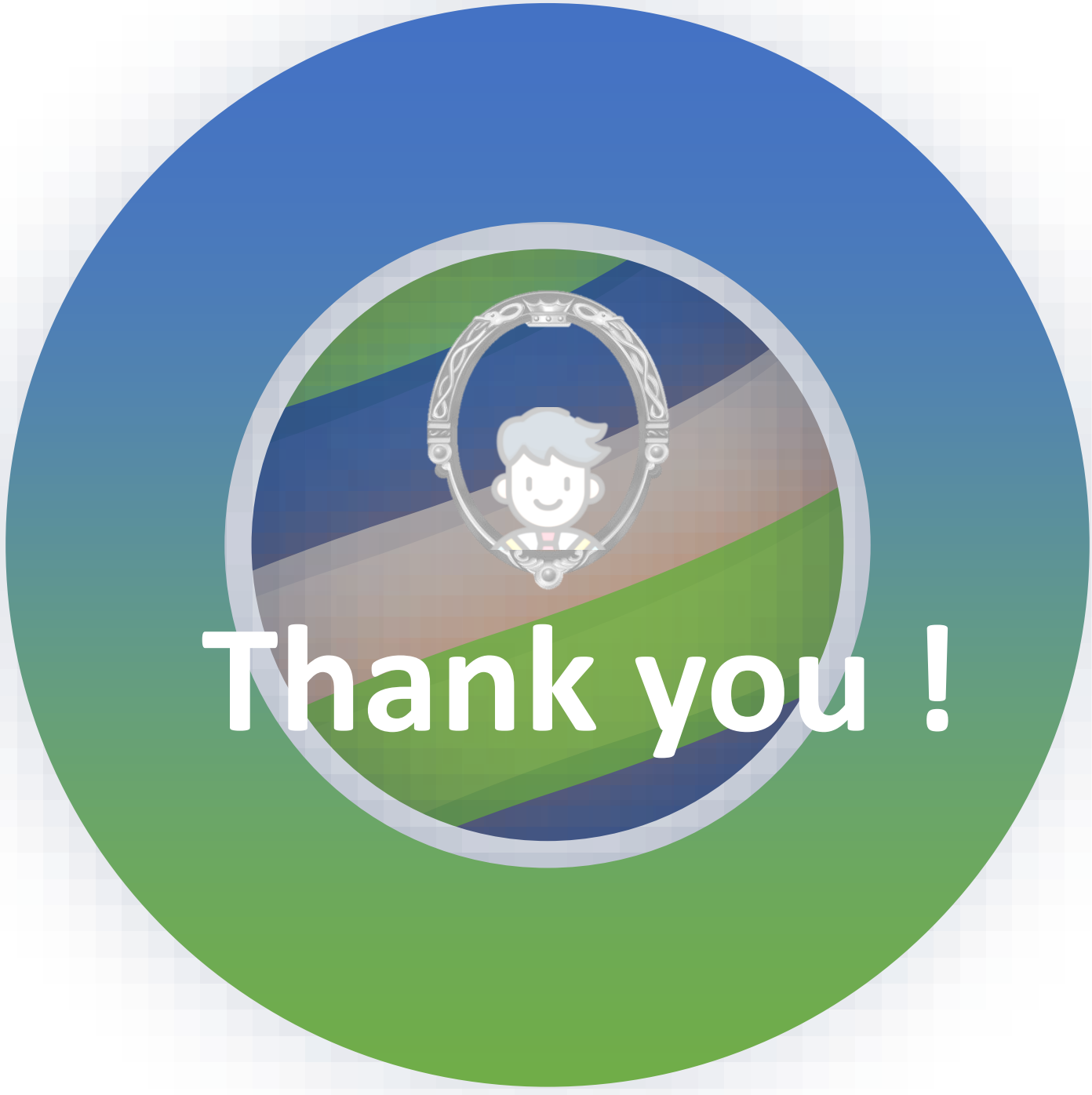


Best method
selection strategy



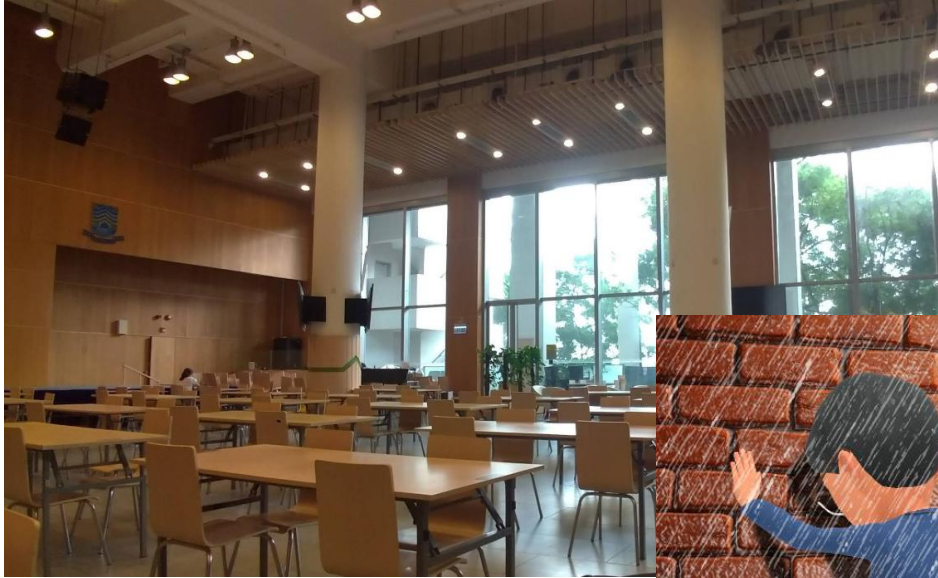
Name	Date modified	Type	Size
RF	9/1/2023 14:19	Python file	1 KB
RF_more	9/1/2023 14:19	Python file	1 KB
data_to_predict	9/1/2023 11:01	Microsoft Excel C...	14 KB
data_to_predict_with_more_columns	9/1/2023 14:19	Microsoft Excel C...	17 KB
digitization	9/1/2023 14:19	Python file	21 KB
digitization_more	9/1/2023 14:19	Python file	21 KB
input information	9/1/2023 14:19	Microsoft Excel W...	15 KB
input_data	9/1/2023 14:19	Microsoft Excel C...	10 KB
input_data_with_more_columns	9/1/2023 14:19	Microsoft Excel C...	10 KB

Sample data and
sample codes



Thank you !

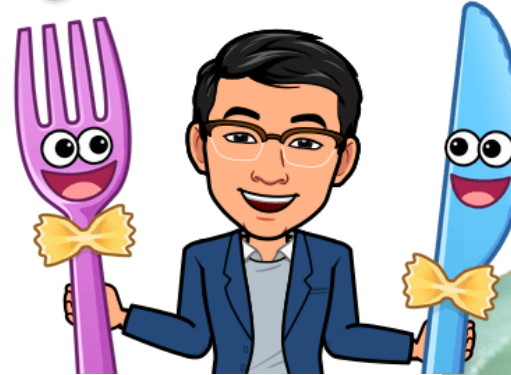
Food at CUHK



Food in CUHK



LET'S EAT!



Toilets at CUHK

